	Krpqlx	Cmfbss
Krpqlx	-1,-1	-12,0
Cmfbss	0,-12	-8,-8

Game Theory 101: The Basics

By William Spaniel

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Please report possible errors to williamspaniel@gmail.com. Since this is an electronic book, I will issue corrections whenever appropriate.

Table of Contents

Lesson 1.1: The Prisoner’s Dilemma and Strict Dominance
Lesson 1.2: Iterated Elimination of Strictly Dominated Strategies
Lesson 1.3: The Stag Hunt, Pure Strategy Nash Equilibria, and Best Responses
Lesson 1.4: Dominance and Nash Equilibrium
Lesson 1.5: Matching Pennies and Mixed Strategy Nash Equilibrium
Lesson 1.6: Calculating Payoffs
Lesson 1.7: Strictly Dominant Mixed Strategies
Lesson 1.8: The Odd Rule and Infinitely Many Equilibria
Final Thought
About the Author

Lesson 1.1: The Prisoner's Dilemma and Strict Dominance

The prisoner's dilemma is the oldest and most studied game in game theory, and its solution concept is also the simplest. As such, we will start with it.

Two thieves plan to rob an electronics store. As they approach the backdoor, the police arrest them for trespassing. The cops suspect that the pair had planned to break in but lack the evidence to support such an accusation. They therefore require a confession to charge the suspects with the greater crime.

Having studied game theory in college, the interrogator throws them into the prisoner's dilemma. He privately sequesters both robbers and tells them the following:

We are currently charging you with trespassing, which calls for a one month jail sentence. I know you were planning on robbing the store, but right now I cannot prove it—I need your testimony. In exchange for your cooperation, I will dismiss your trespassing charge, and your partner will be charged to the fullest extent of the law—a twelve month jail sentence.

I am offering your partner the same deal. If both of you confess, your individual testimony is no longer as valuable, and your jail sentence will be eight months each.

If both criminals are self-interested and only care about minimizing their jail time, should they take the interrogator's deal?

Solving the Game

The story contains a lot of information. Luckily, we can condense everything we need to know into a simple matrix:

	Keep Quiet	Confess
Keep Quiet	-1, -1	-12, 0
Confess	0, -12	-8, -8

We will use this type of game matrix regularly, so it is important to understand what it means. There are two players in this game. The first player's moves ("keep quiet" and "confess") are in the rows, and the second player's moves are in the columns; the first player's payoffs are listed first for each outcome, and the second player's are listed second. For example, if the first player keeps quiet and the second player confesses, then the game ends in the top right set of payoffs, with the first player receiving twelve months of jail time and the second player receiving zero. Finally, as a matter of convention, we refer to the first player as a man and the second player as a woman; this will allow us to utilize pronouns like "he" and "she" instead of repeating "player 1" and "player 2" *ad nauseam*.

So what is each player's optimal strategy? To see the answer, it is best to look at each move in isolation. Look at this from the perspective of player 1. Suppose

he knew player 2 would keep quiet. What should he do? Let's check the important information in that context and replace everything that does not matter with a question mark:

	Keep Quiet
Keep Quiet	-1, ?
Confess	0, ?

From this, it is clear that player 1 should confess. If he keeps quiet, he will spend one month in jail. But if he confesses, he walks away. Since he prefers less jail time to more jail time, he should confess in this situation. Note that player 2's payoffs are completely irrelevant to player 1's decision in this context—if he knows that she will keep quiet, then he only needs to look at his own payoffs to decide which strategy to pick. Thus, the question marks could be any number at all, and player 1's decision will remain the same.

Now suppose player 1 knew that player 2 would confess. What should he do? Again, the answer is easier to see if we look at the only information that matters here:

	Confess
Keep Quiet	-12, ?
Confess	-8, ?

Confession wins a second time: confessing leads to eight months of jail time, whereas silence buys twelve. So player 1 would want to confess if player 2 confesses.

Putting these two pieces of information together, we reach an important conclusion—player 1 is better off confessing regardless of what player 2 does! Thus, player 1 can effectively ignore whatever he thinks player 2 will do, since confessing gives him less jail time in either scenario.

Now let's take player 2's perspective. Suppose she knew that player 1 will keep quiet, even though we realize he should not. Here is her situation:

	Keep Quiet	Confess
Keep Quiet	?, -1	?, 0

As before, player 2 should confess, as he will shave a month off her jail sentence for doing so.

Finally, suppose she knew player 1 will confess. How should she respond?

	Keep Quiet	Confess
Confess	?, -12	?, -8

Unsurprisingly, she should confess and spend four fewer months in jail.

Once more, player 2 is better off confessing regardless of what player 1 does. Thus, we have reached a solution: both players confess, and both players spend eight months in jail. The justice system has triumphed, thanks to the interrogator's savviness!

This outcome perplexes a lot of people new to the field of game theory. Looking at the game matrix, they see that the <keep quiet, keep quiet> outcome leaves both players better off than the <confess, confess> outcome and wonder why the players cannot coordinate on the more efficient outcome. But as we just saw, any promise to keep quiet is unbelievable. Player 1 *wants* player 2 to keep quiet so when he confesses he walks away freely. The same goes for player 2. As a result, the <keep quiet, keep quiet> outcome is inherently unstable, which is why the players ultimately end up in the inferior (but sustainable) <confess, confess> outcome.

Meaning of the Numbers

Although there is a large branch of game theory devoted to the study of expected utility, we generally consider each player's payoffs as a ranking of his most preferred outcome to his least preferred outcome. In the prisoner's dilemma, we *assumed* that players only wanted to minimize their jail time. Game theory does *not* force players to believe that is the case, as critics frequently claim. Instead, game theory analyzes what should happen given what players desire. So if players only want to minimize jail time, we could use the negative number of years spent in jail as their payoffs. This preserves their individual orderings over outcomes, as the most preferred outcome is worth 0, the least preferred outcome is -10, and everything else logically follows in between.

Interestingly, the cardinal values of the numbers are irrelevant to the outcome of the game. For example, suppose we changed the matrix to this:

	Keep Quiet	Confess
Keep Quiet	3, 3	1, 4
Confess	4, 1	2, 2

Here, we have replaced the months of jail time with an ordering of most to least preferred outcomes, with 4 representing a player's most preferred outcome and 1 representing a player's least preferred outcome. In other words, player 1 would most like to reach the <confess, keep quiet> outcome, then the <keep quiet, keep quiet> outcome, then the <confess, confess> outcome, then the <keep quiet, confess> outcome.

Given that, confess still strictly dominates keep quiet. To see this, suppose player 2 kept quiet:

	Keep Quiet
Keep Quiet	3, ?
Confess	4, ?

Player 1 should confess, since 4 is better than 3.

Likewise, suppose player 2 confessed:

	Confess
Keep Quiet	1, ?
Confess	2, ?

Then player 1 should still confess, as 2 is better than 1.

The same is true for player 2. First, suppose player 1 kept quiet:

	Keep Quiet	Confess
Keep Quiet	?, 3	?, 4

Player 2 ought to confess, since 4 is greater than 3.

Alternatively, if player 1 confessed:

	Keep Quiet	Confess
Confess	?, 1	?, 2

Player 2 should confess as well, as 2 is greater than 1. Thus, regardless of what the other player does, each player's best strategy is to confess.

To be clear, this preference ordering exclusively over time spent in jail is just one way the players may interpret the situation. Suppose you and a friend were actually arrested and the interrogator offered you a similar deal. The results here do not generally tell you what to do in that situation, unless you and your friend only cared about jail time. In contrast, perhaps your friendship is strong, and both of you value it more than avoiding jail time. Since confessing might destroy friendship, you could prefer to keep quiet if your partner kept quiet, which changes the ranking of your outcomes. Your preferences here are perfectly rational. However, we do not yet have the tools to solve the corresponding game. We will reconsider these alternative sets of preferences in lesson 1.3.

Nevertheless, for the given payoffs, we have seen an example of *strict dominance*. We say that a strategy X strictly dominates strategy Y for a player if strategy X provides a greater payoff for that player than strategy Y regardless of what the other players do. In this example, confess strictly dominated keep quiet for both players. Unsurprisingly, intelligent players never play strictly dominated strategies—by definition, there is *always* a better option regardless of what the other players do.

Deadlock

We can solve many games with strict dominance alone. Take this game, for example:

	Left	Right
Up	3, 3	4, 1
Down	1, 4	2, 2

Here, up strictly dominates down for player 1 and left strictly dominates right for player 2. We can verify this using the same process as before. First, suppose

player 2 chooses left:

	Left
Up	3, ?
Down	1, ?

We see that up is better than down, as 3 is greater than 1. Repeating this for right, we focus on the following:

	Right
Up	4, ?
Down	2, ?

Once more, up is better than down, since 4 is greater than 2. So up is a better strategy than down regardless of what player 2 does.

Switching gears, suppose player 1 selected up. Then player 2 can focus on the following contingency:

	Left	Right
Up	?, 3	?, 1

It is clear that left is better than right in this case, as 3 is greater than 1. Repeating this process a final time, player 2 now assumes player 1 will play down:

	Left	Right
Down	?, 4	?, 2

Player 2 sees that left is still better than right, as 2 is greater than 1. Since left always beats right regardless of what player 1 does, left strictly dominates right, and therefore player 2 will play left. Thus, the outcome is <up, left>.

The technical name for this game is “deadlock”; both players are locked into their strictly dominant strategy and will never achieve their best outcome unless the other makes a mistake. However, unlike in the prisoner’s dilemma, there is no alternative outcome that is simultaneously better for both players than the <up, left> solution. As such, this game is more intuitive but substantively less interesting.

Overall, strict dominance is a powerful tool in game theory. But while the concept is simple, applying it can be difficult. Even in matrix form, a game still has a lot of information. To successfully find dominated strategies, we must focus on one player’s payoffs at a time. Above, we used question marks to isolate the relevant payoffs. When searching for strictly dominated strategies on your own, be sure to mentally block out the irrelevant payoffs and strategies in a similar manner.

Lesson 1.2: Iterated Elimination of Strictly Dominated Strategies

Consider the following game:

	Left	Center	Right
Up	13, 3	1, 4	7, 3
Middle	4, 1	3, 3	6, 2
Down	-1, 9	2, 8	8, -1

This game presents a new challenge. With the prisoner's dilemma, we knew exactly what each player would do—confessing was always better than remaining silent regardless of what the other player's strategy. There is no similar simplicity with this game. Indeed, player 1 is in a quandary. Suppose he knew player 2 would go left:

	Left
Up	13, ?
Middle	4, ?
Down	-1, ?

Player 1 would then want play up, as the 13 points he earns there are better than the 4 he earns for middle and -1 he earns for down. But this is not the case if player 2 chose center:

	Center
Up	1, ?
Middle	3, ?
Down	2, ?

Now player 1 ought to play middle, as those 3 points are better than the 1 point for going up and the 2 points for going down. And to make things more complicated, look what happens if player 1 knew player 2 would choose right:

	Right
Up	7, ?
Middle	6, ?
Down	8, ?

In this case, player 1 should go down and earn 8, which is superior to up and earning 7 or middle and earning 6.

So in three different cases, player 1 should do three different things. How does player 1 decide which to choose?

<u>Iterated Elimination of Strictly Dominated Strategies</u>

In the previous lesson, we discussed why players ought to never play strictly dominated strategies. In game theory, we hold the players to a high standard—namely, that they maximize their payoffs and know that the other player is trying to do the same. This has important implications on how they ought to interact.

With that in mind, let's look back at the prisoner's dilemma:

	Keep Quiet	Confess
Keep Quiet	-1, -1	-12, 0
Confess	0, -12	-8, -8

Recall that confess strictly dominates keep quiet for player 1. That is, if player 2 were to keep quiet, then player 1 would want to confess (0 is better than -1); and if player 2 were to confess, then player 1 would want to confess (-8 is better than -12). So player 1 would never keep quiet.

Now switch to player 2's perspective. She knows that player 1 is intelligent enough to see that confess strictly dominates keep quiet. Therefore, she infers that he will never keep quiet. Given that, for all intents and purposes, player 2 can ignore keep quiet as a strategy for player 1—he will never play it. As such, from her perspective, she can reduce the game to the following:

	Keep Quiet	Confess
Confess	0, -12	-8, -8

At this point, it is obvious that player 2 should confess and earn -8 rather than keep quiet and earn -12. Note that she never even had to observe that confess strictly dominated keep quiet for her as well; instead, she merely reduced the game to the plausible choices for player 1. And since the only choice that is left for player 1 is to confess, player 2 can condition her response purely based off of that information.

This process is known as iterated elimination of strictly dominated strategies (IESDS). We began by noting that confess strictly dominated keep quiet for player 1, so we erased keep quiet as a strategy for him. We then pretended that the remaining game was all that mattered, and found that confess strictly dominated keep quiet for player 2.

IESDS allows us to take some fairly complicated games and reduce them into smaller games that are easily solved, such as the original game of this lesson:

	Left	Center	Right
Up	13, 3	1, 4	7, 3
Middle	4, 1	3, 3	6, 2
Down	-1, 9	2, 8	8, -1

Let's isolate player 2's choice between center and right:

	Center	Right
Up	?, 4	?, 3
Middle	?, 3	?, 2
Down	?, 8	?, -1

Should player 2 ever play right? She should not—center strictly dominates right. To see this, note that if player 1 plays up, then center is better than right, as 4 is greater than 3. Likewise, if he plays middle, center's 3 is superior to right's 2. Finally, if player 1 plays down, center beats right 8 to -1. Consequently, regardless of what player 1 does, center is better than right for player 2.

Now look at the game from player 1's perspective. He knows that player 2 will never play right, so he only needs to consider the following game:

	Left	Center
Up	13, 3	1, 4
Middle	4, 1	3, 3
Down	-1, 9	2, 8

Let's focus on player 1's decision between middle and down:

	Left	Center
Middle	4, ?	3, ?
Down	-1, ?	2, ?

Down is no longer a sensible choice—middle strictly dominates it, as 4 is greater than -1 and 3 is greater than 2. Essentially, down's usefulness was a contingency plan against player 2 playing right. But since right is not a sensible choice for player 2, she will not play it. And since player 1 knows that she is sensible and will not play right, there is no reason for him to play down.

Let's flip back to player 2's perspective. She knows that right is a lousy choice for her. She also knows player 1 is aware of the fact that she knows this. As a result, player 1 infers that down is a lousy choice for him, and she is aware of the fact he infers this. Working through that logic, she knows he will not play down, and therefore she can reduce the original game to the following:

	Left	Center
Up	13, 3	1, 4
Middle	4, 1	3, 3

Center now strictly dominates left for her:

	Left	Center
Up	?, 3	?, 4
Middle	?, 1	?, 3

We see that if player 1 plays up, then center is better than left, as 4 is greater than 3. And likewise, if player 1 plays middle, center is again better than left because 3 is greater than 1. So player 2 would never play left.

After working through another level of the he-knows-that-she-knows-that-he-knows logic, player 1 can conclude he is looking at the following reduced game:

	Center
Up	1, 4
Middle	3, 3

Isolating player 1's payoffs makes it clear that he will want to play middle since 3 is greater than 1:

	Center
Up	1, ?
Middle	3, ?

Therefore, we can conclude that player 1 will play middle and player 2 will

play center.

Does Order Matter?

Suppose we had a game that started with two strictly dominated strategies. A natural question is whether we will end up with a different answer depending on which one we choose to eliminate first.

In fact, our first choice is irrelevant. The reason for this is a little complicated, so let's instead look at a couple of examples. Earlier in this section, we solved the prisoner's dilemma by eliminating player 1's keep quiet strategy first. Based off of the remaining game, we eliminated keep quiet for player 2. Thus, the solution was for both players to confess.

However, since we know that keep quiet strictly dominates confess for player 2 in the original game, we could have started by removing her keep quiet strategy first. That would have left us with this game:

	Confess
Keep Quiet	-12, 0
Confess	-8, -8

Unsurprisingly, confess strictly dominates keep quiet in the remaining game for player 1:

	Confess
Keep Quiet	-12, ?
Confess	-8, ?

Since -8 is greater than -12, player 1 will confess. Thus, both players confess.

Let's move on to a more challenging example:

	Left	Right
Up	-1, 1	4, 2
Middle	0, 2	3, 3
Down	-2, -2	2, -1

We might begin by noting that middle strictly dominates down for player 1:

	Left	Right
Middle	0, ?	3, ?
Down	-2, ?	2, ?

That is, 0 is greater than -2 and 3 is greater than 2. So player 1 would never want to play down, which reduces the original game to the following:

	Left	Right
Up	-1, 1	4, 2
Middle	0, 2	3, 3

Now we can see that right strictly dominates left for player 2:

	Left	Right
Up	?, 1	?, 2
Middle	?, 2	?, 3

This is because 2 is greater than 1 and 3 is greater than 2. Thus, we can

eliminate left from the game:

	Right
Up	4, 2
Middle	3, 3

And now the game boils down to what is best for player 1:

	Right
Up	4, ?
Middle	3, ?

Since 4 is greater than 3, player 1 will play up. Thus, the solution is <up, right>.

Let's go back to the original game:

	Left	Right
Up	-1, 1	4, 2
Middle	0, 2	3, 3
Down	-2, -2	2, -1

While middle strictly dominates down, an astute observer will note that right also strictly dominates left in this game:

	Left	Right
Up	?, 1	?, 2
Middle	?, 2	?, 3
Down	?, -2	?, -1

That is, 2 is greater than 1, 3 is greater than 2, and -1 is greater than -2. So instead of beginning by eliminating down, we could have begun by eliminating left instead. Had we done so, our next step would have been the following:

	Right
Up	4, 2
Middle	3, 3
Down	2, -1

From here, it is only a matter of selecting player 1's greatest payoff:

	Right
Up	4, ?
Middle	3, ?
Down	2, ?

Since 4 is greater than 3 or 2, player 1 plays up. Therefore, we end up at the <up, right> outcome regardless of which path we take. This holds for *any* game when we use iterated elimination of strictly dominated strategies. Consequently, when you are solving complex games and you find a strictly dominated strategy, *eliminate it immediately*. Although there may be more strategies you could eliminate in the first step, these strategies will still be strictly dominated in the next step. It will also be easier to find them, as there is less information to consider in the remaining game.

Weak Dominance

We must be careful when we use iterated elimination of *strictly* dominated strategies. To illustrate a potential pitfall, consider this game:

	Left	Right
Up	0, 1	-4, 2
Middle	0, 3	3, 3
Down	-2, 2	3, -1

Let's first focus on player 1's choice between up and middle:

	Left	Right
Up	0, ?	-4, ?
Middle	0, ?	3, ?

You might be tempted to say that middle strictly dominates up for player 1. However, that is not the case. If player 2 plays right, then middle is better than up, as 3 is greater than -4. But if player 2 plays left, player 1 is indifferent between up and middle; regardless of which he chooses, he will wind up with a payoff of 0. Strict dominance requires middle to always be better than up; equally good does not cut it. Instead, we say that middle *weakly* dominates up. In general, we say a strategy X weakly dominates a strategy Y for a player if X provides *at least* as great of a payoff for that player regardless of what the other players do, and there is at least one set of opposing strategies for which X pays greater than Y.

Eliminating weakly dominated strategies and analyzing the remaining game is called iterated elimination of weakly dominated strategies (IEWDS). Depending on the game, IEWDS sometimes produces sensible answers, and sometimes it does not. Unfortunately, we do not know what type of game we are looking at simply by eliminating weakly dominated strategies.

To fully see the problem, let's assume we could use IEWDS the same way we use IESDS. Since middle weakly dominated up, let's eliminate it and see what happens:

	Left	Right
Middle	0, 3	3, 3
Down	-2, 2	3, -1

We now see that left weakly dominates right for player 2:

	Left	Right
Middle	?, 3	?, 3
Down	?, 2	?, -1

That is, 3 is equal to 3 and 2 is greater than -1. If we eliminate the weakly dominated right, we are left with:

	Left
Middle	0, 3
Down	-2, 2

And since player 1 earns 0 for middle and -2 for down, he will choose middle. Thus, we conclude that the outcome of this game is <middle, left>. But is it?

Let's start over:

	Left	Right
Up	0, 1	-4, 2
Middle	0, 3	3, 3
Down	-2, 2	3, -1

We began last time by observing that middle weakly dominates up. However, middle also weakly dominates down:

	Left	Right
Middle	0, ?	3, ?
Down	-2, ?	3, ?

That is, 0 is greater than -2 and 3 is equal to 3. Eliminating down yields:

	Left	Right
Up	0, 1	-4, 2
Middle	0, 3	3, 3

Now we see that right weakly dominates left for player 2:

	Left	Right
Up	?, 1	?, 2
Middle	?, 3	?, 3

That is, 2 is greater than 1 and 3 is equal to 3. So eliminating left leaves us with:

	Right
Up	-4, 2
Middle	3, 3

At this point, player 1 picks the strategy that earns him the highest payoff. Since 3 is greater than -4, he selects middle, leaving us with a solution of <middle, right>. Yet earlier we said the solution was <middle, left>. The problem is that iterated elimination of weakly dominated strategies does not give us any guidance about which is correct or if both are. To resolve the issue, we need to introduce a broader solution concept. We will do that in the next lesson.

Lesson 1.3: The Stag Hunt, Pure Strategy Nash Equilibrium, and Best Responses

Two hunters enter a range filled with hares and a single stag. Hares are unintelligent and easy to capture. The stag, on the other hand, is cunning and can only be caught if the hunters work together. Without any communication, the hunters independently choose whether to hunt hares or the stag. If they both hunt hares, they each capture half of the hares in the range. If one hunts the stag and the other hunts hares, the stag hunter goes home empty-handed while the hare hunter captures all of the hares. Finally, if both hunt the stag, then each of their shares of the stag is greater than the value of all of the hares.

The following matrix depicts the strategic situation:

	Stag	Hare
Stag	3, 3	0, 2
Hare	2, 0	1, 1

So far, we only know how to solve games using iterated elimination of strictly dominated strategies. Let's search for them here.

First, suppose player 1 knew that player 2 will hunt a stag:

	Stag
Stag	3, ?
Hare	2, ?

In this case, hunting a stag is optimal for player 1 as well: doing so nets him 3 points, whereas chasing hares gives him only 2.

Now suppose player 1 knew player 2 will hunt hares:

	Hare
Stag	0, ?
Hare	1, ?

Hunting a stag is no longer optimal for player 1; hare has stag beat by a 1 to 0 margin. Thus, player 1 has no strictly (or weakly) dominated strategy. In fact, player 1's optimal strategy is completely dependent on what player 2 does. If she hunts a stag, so should he; but if she hunts for hares, he ought to as well.

Given the symmetry of the game, the same is true for player 2: she should also do whatever player 1 does. To quickly verify this, suppose player 1 hunted a stag:

	Stag	Hare
Stag	?, 3	?, 2

Again, player 2 should choose stag, since 3 is greater than 2. But say player 1 hunted for hares:

	Stag	Hare
Hare	?, 0	?, 1

Then she ought to opt for hares, as 1 is greater than 0.

How do we solve games that do not have strictly dominated strategies? We look for Nash equilibria. A Nash equilibrium is a set of strategies, one for each player, such that no player has incentive to change his or her strategy given what the other players are doing.

Some examples will clarify that definition. First, consider the set of strategies <stag, stag>. Does either player have incentive to change his or her strategy?

Let's look at this from the perspective of player 1. First, we have to hold player 2's strategy constant; that is, we are assuming that player 2 will stick to her strategy of stag. Should player 1 switch his strategy?

	Stag
Stag	3, ?
Hare	2, ?

We just saw this image, so we know he should not: 3 is still greater than 2.

What about player 2? Similarly, we must hold player 1's strategy constant and ask ourselves whether player 2 would want to change her strategy:

	Stag	Hare
Stag	?, 3	?, 2

Once more, she would not want to: 3 remains greater than 2. So <stag, stag> is a Nash equilibrium. Specifically, we call it a pure strategy Nash equilibrium (PSNE) because both players are playing deterministic strategies. That is, in this equilibrium, player 1 *always* plays stag and player 2 *always* plays stag. We will only focus on PSNE in this section. (In contrast, lesson 1.5 covers mixed strategy Nash equilibrium, or MSNE, in which players randomize between their strategies. For example, a mixed strategy might be to flip a coin and hunt hares on heads and the stag on tails.)

Are there any other PSNE? Let's start by seeing if player 1 would want to switch his strategy in the <stag, hare> outcome:

	Hare
Stag	0, ?
Hare	1, ?

He should alter his strategy. If he keeps hunting a stag, he will end up with 0. But if he switches his strategy to hare, he can profitably deviate to 1. If even a single player would want to deviate, a set of strategies is not a Nash equilibrium. So without even checking player 2's move, we can throw out <stag, hare>. But let's check anyway for the sake of practice:

	Stag	Hare
Stag	?, 3	?, 2

This is the third time we have seen this image, so by now we should know player 2 has a profitable deviation: she should switch to stag.

Now let's check whether <hare, stag> is a Nash equilibrium. Given the

game's symmetry and how <stag, hare> is not a Nash equilibrium, it should be obvious that <hare, stag> is not either. To wit, consider player 1's choice:

	Stag
Stag	3, ?
Hare	2, ?

Currently, he is making 2; if he switches to stag, he will get 3. Since that is a profitable deviation, <hare, stag> is not a Nash equilibrium.

Once more, we could have also verified that <hare, stag> is not a Nash equilibrium by looking at player 2's choice:

	Stag	Hare
Hare	?, 0	?, 1

Optimally, player 2 should switch from hunting a stag and earning 0 to chasing hares and earning 1.

Finally, let's check whether <hare, hare> is a Nash equilibrium. We will begin with player 1's choice:

	Hare
Stag	0, ?
Hare	1, ?

Hare remains optimal for player 1; switching to stag changes his payoff from 1 to 0. So the only way for <hare, hare> to not be a Nash equilibrium is if player 2 would want to switch. Let's check if that is the case:

	Stag	Hare
Hare	?, 0	?, 1

She should not switch—deviating also decreases her payoff from 1 to 0. Since neither player has incentive to change his or her strategy, <hare, hare> is a Nash equilibrium. Therefore, the stag hunt as two pure strategy Nash equilibria: <stag, stag> and <hare, hare>.

Unlike the prisoner's dilemma, the stag hunt illustrates game theory's power to analyze interdependent decision making. With the prisoner's dilemma, each could effectively ignore what the other one planned on doing since confess generated a strictly greater payoff regardless of the other prisoner's choice. That is not the case with the stag hunt. Here, each player wants to do what the other is doing. That is, each player's individually optimal strategy is entirely a function of the other player's choice.

The stag hunt also highlights how Nash equilibria do not have to be efficient. Indeed, both players are better off with the <stag, stag> equilibrium than the <hare, hare> equilibrium—and, unlike the prisoner's dilemma, both outcomes are sustainable, because neither player has incentive to change his or her strategy given what the other is doing. As such, although the players might want to coordinate on the <stag, stag> outcome, they might get stuck in the inefficient

<hare, hare> outcome.

To see how this is plausible, suppose both players saw a sign that said “today is hare hunting day” on their way to the hunting range. What should the hunters do? Unfortunately, given the nature of the game, there is no clear answer. It is in both hunters’ best interest to ignore the sign and hunt a stag, as long as both do that. But if I think that you are going to follow the sign and hunt hares, then I should follow the sign as well, even if I really want to hunt a stag. And if you think I am going to think that you are going to follow the sign, even if you have no plans on doing so, you should still hunt hares because you anticipate that I will as well. Simple coordination—a cell phone call from one hunter to the other—solves the issue. But absent that, both outcomes are plausible, which is one of the motivations behind Nash equilibria.

As a final note, Nash equilibrium only looks at *individual* deviations. That is, we need to check whether individuals cannot individually deviate to better payoffs. If both players are playing hare, then there is a collectively profitable deviation to both playing stag. But individually, both are better off staying with their hare strategy. As such, it is a Nash equilibrium.

New Preferences in the Prisoner’s Dilemma

Lesson 1.1 claimed that if players did solely want to minimize jail time, their optimal behaviors might change. Indeed, game theory was *not* making a normative claim that jail time is the only thing that players should about; instead, we were seeking the optimal behaviors of the prisoners given that they had such preferences.

With that in mind, let’s reframe the game. Suppose both prisoners were good friends and would rather keep quiet if they knew that the other would as well. In this case, each player’s most preferred outcome is <keep quiet, keep quiet>, then <confess, keep quiet>, then <confess, confess>, then <keep quiet, confess>. Expected utility values of 3, 2, 1, and 0 preserve the rankings of these outcomes. Consequently, we could draw the payoff matrix like this:

	Keep Quiet	Confess
Keep Quiet	3, 3	0, 2
Confess	2, 0	1, 1

Notice that these are the exact same payoffs that we saw in the stag hunt. As such, there are two PSNE: <keep quiet, keep quiet> and <confess, confess>. Now the interrogator’s plan may fail, as the players have a strong enough relationship that their preferences allow for mutual cooperation. That is, <keep quiet, keep quiet> is a sustainable outcome in this version of the prisoner’s

dilemma, as neither player has incentive to deviate from that set of strategies.

There are two things to take away from this reworking of the prisoner's dilemma. First, it shows how outcomes are a function of preferences and not just the strategic environment. The interrogator made the same offer in both versions of the game. In the original version, he induced both to confess. But when the players had friendlier preferences, his plot could very well fail, provided the prisoners can coordinate on the <keep quiet, keep quiet> equilibrium.

Second, it shows game theory's ability to draw parallels between seemingly dissimilar situations. In this section alone, we have looked at two completely different scenarios. One dealt with hunting; the other dealt with two friends in a legal predicament. Yet, once we strip down all of the irrelevant features of the strategic interaction, we see that a single game underlies both cases. In this way, game theory allows us to connect seemingly disparate situations under a common framework.

Best Responses

Two generals each have three units and are preparing for an upcoming battle. Each can choose to send any number of units to the fight or none at all. The side with more troops wins the battle, and the fight will draw if there are equal forces. Victory is worth 1 point; defeat is worth -1. If the sides draw or at least one side refuses to fight, both sides earn 0.

This is a "safety in numbers" game, and the following matrix represents the generals' situation:

	0	1	2	3
0	0, 0	0, 0	0, 0	0, 0
1	0, 0	0, 0	-1, 1	-1, 1
2	0, 0	1, -1	0, 0	-1, 1
3	0, 0	1, -1	1, -1	0, 0

When we found all the pure strategy Nash equilibria in the stag hunt, we went through every outcome one at a time and checked whether a player had a profitable deviation. That was a reasonable task when there were only four outcomes, but safety in numbers has sixteen. We could go through each of the sixteen outcomes, but that would be time consuming. Instead, we will use a different method that involves marking best responses.

A best response is simply the optimal strategy for a particular player given what everyone else is doing. For example, suppose general 2 was going to send 0 units to the battle. What is general 1's best response?

	0
0	0, ?
1	0, ?
2	0, ?
3	0, ?

As the image demonstrates, it really does not matter what strategy general 1 picks—the battle never takes place, so every strategy generates a payoff of 0 for him. As such, all of these strategies are best responses for general 1. For book keeping purposes, we mark a player's best responses with an asterisk over his or her payoffs, like this:

	0
0	0*, ?
1	0*, ?
2	0*, ?
3	0*, ?

We will see why these asterisks come in handy in a moment.

Now suppose general 2 will send one unit to the battle. Let's mark the best responses:

	1
0	0, ?
1	0, ?
2	1*, ?
3	1*, ?

There are two best responses: sending two and sending three units. Sending zero units results in no battle taking place and a payoff of 0. Sending one unit results in a draw and a payoff of 0. Sending two or sending three units gives general 1 the victory and a payoff of 1. Since 1 is the greatest payoff, we mark two and three as best responses.

Next, suppose general 2 sent two units to the battle:

	2
0	0, ?
1	-1, ?
2	0, ?
3	1*, ?

Here, sending one loses the battle, sending two leads to a draw, and sending three earns the win. Since the payoff of 1 is the greatest of all the outcomes, general 1's best response is to send three units. As such, only three gets an asterisk.

Lastly, suppose general 2 sent three units:

	3
0	0*, ?
1	-1, ?
2	-1, ?
3	0*, ?

Now there are two best responses. If general 1 sends one or two units, he loses the battle and earns -1. If he sends no one, the battle does not take place, and he gets 0. And if he sends everyone, the battle draws, and he also receives a payoff of 0. Both of the 0 outcomes are the best, so both of them earn asterisks.

Having marked all of general 1's best responses, the game looks like this:

	0	1	2	3
0	0*, 0	0, 0	0, 0	0*, 0
1	0*, 0	0, 0	-1, 1	-1, 1
2	0*, 0	1*, -1	0, 0	-1, 1
3	0*, 0	1*, -1	1*, -1	0*, 0

However, we are only half way done—we still need to mark general 2's best responses. As before, let's start by assuming general 1 will send zero units:

	0	1	2	3
0	?, 0*	?, 0*	?, 0*	?, 0*

We see that every strategy is a best response for general 2 if general 1 sends zero units.

Now check what happens if general 2 sends one unit:

	0	1	2	3
1	?, 0	?, 0	?, 1*	?, 1*

Sending two or three units nets 1. Therefore, two and three are best responses.

Next, let's consider general 2's best responses to general 1 sending two units:

	0	1	2	3
2	?, 0	?, -1	?, 0	?, 1*

Sending three units nets 1 point, whereas the other alternatives are worth 0 or -1. So sending three units is the only best response here.

Finally, suppose general 1 sent three units:

	0	1	2	3
3	?, 0*	?, -1	?, -1	?, 0*

Now one and two are losing strategies, while zero and three break even at 0 points of utility. So zero and three are best responses.

Combining all of that information together, the full game looks like this:

	0	1	2	3
0	0*, 0*	0, 0*	0, 0*	0*, 0*
1	0*, 0	0, 0	-1, 1*	-1, 1*
2	0*, 0	1*, -1	0, 0	-1, 1*
3	0*, 0*	1*, -1	1*, -1	0*, 0*

Was all of that work worth the effort? Most definitely. To find all of the game's pure strategy Nash equilibria, we only need to check which outcomes have asterisks next to both players' payoffs. We see there are four such outcomes: <0, 0>, <0, 3>, <3, 0>, and <3, 3>. Since the best responses method is the most efficient way of finding pure strategy Nash equilibria, we will be relying on it in upcoming lessons. Consequently, if you do not yet have the hang of it, there will be plenty more practice ahead.

Lesson 1.4: Dominance and Nash Equilibrium

This lesson investigates the relationship between Nash equilibrium and dominance. We will see two things. First, if iterated elimination of strictly dominated strategies reduces the game to a single outcome, that outcome is a Nash equilibrium and it is the *only* Nash equilibrium of that game. And second, iterated elimination of weakly dominated strategies is not as kind: while any solution found through IEWDS is a Nash equilibrium, the IEWDS process sometimes eliminates other Nash equilibria.

Nash Equilibrium and Iterated Elimination of Strictly Dominated Strategies

How does Nash equilibrium relate to iterated elimination of strictly dominated strategies? As it turns out, solutions found through the strict dominance solution concept are Nash equilibria. Consequently, not only is <confess, confess> the dominance solution to the prisoner's dilemma from lesson 1.1, but it is also the Nash equilibrium of that game.

Let's consider a tougher example. Here is the game from lesson 1.2 we used to introduce iterated elimination of strictly dominated strategies:

	Left	Center	Right
Up	13, 3	1, 4	7, 3
Middle	4, 1	3, 3	6, 2
Down	-1, 9	2, 8	8, -1

Through IESDS, we know the solution is <middle, center>. Let's verify that it is also the unique pure strategy Nash equilibrium of the game.

First, let's find player 1's best response to left:

	Left
Up	13*, ?
Middle	4, ?
Down	-1, ?

Up earns the asterisk, as 13 is the largest payoff.

Moving on, suppose player 2 choose center:

	Center
Up	1, ?
Middle	3*, ?
Down	2, ?

Middle gets the asterisk this time, as 3 is now the largest payoff for player 1.

Lastly for player 1's best responses, we consider player 2 moving right:

	Right
Up	7, ?
Middle	6, ?
Down	8*, ?

Down earns the asterisk with a payoff of 8.

Switching gears, let's find the best responses for player 2 if player 1 moves

up:

	Left	Center	Right
Up	?, 3	?, 4*	?, 3

With 4 being the largest, player 2's best response is to move center.

Now suppose player 1 moved middle:

	Left	Center	Right
Middle	?, 1	?, 3*	?, 2

Center remains the best response, with 3 being greater than 1 or 2.

Finally, consider how player 2 should reply to down:

	Left	Center	Right
Down	?, 9*	?, 8	?, -1

Player 2's best response has shifted to left.

If we reconstruct the original game with all of the best responses marked, we get this:

	Left	Center	Right
Up	13*, 3	1, 4*	7, 3
Middle	4, 1	3*, 3*	6, 2
Down	-1, 9*	2, 8	8*, -1

The only mutual best response is <middle, center>. As claimed, the solution through iterated elimination of strictly dominated strategies was the only Nash equilibrium of the game.

<u>When IESDS Leaves Multiple Strategies</u>

After eliminating all strictly dominated strategies, you will sometimes find multiple strategies remain available for each player. How do you handle this? We simply fall back to the other techniques to find Nash equilibria.

To illustrate, let's add some spice to the stag hunt. Suppose player 1 could hunt a stag, hares, or player 2. To make things fair to player 2, she could hunt a stag, hunt hares, or hide from player 1. If player 1 goes after his hunting partner and player 2 hides, then player 2 survives. Otherwise, she earns an extremely negative payoff. (She will be shot, after all.) Fortunately for player 2, player 1 will feel guilty if he ever shoots player 2, so he also earns a negative payoff if she is ever hurt.

Adding these new strategies and payoffs leaves us with this game:

	Stag	Hare	Hide
Stag	3, 3	0, 2	0, 0
Hare	2, 0	1, 1	2, 0
Human	-5, -10	-5, -10	0, 0

To solve this game, we should first note that hare strictly dominates human for player 1:

	Stag	Hare	Hide
Hare	2, ?	1, ?	2, ?
Human	-5, ?	-5, ?	0, ?

That is, 2 is greater than -5, 1 is greater than -5, and 2 is greater than 0. Knowing that, we can remove human as a strategy and consider the following:

	Stag	Hare	Hide
Stag	3, 3	0, 2	0, 0
Hare	2, 0	1, 1	2, 0

From here, note that hare strictly dominates hide for player 2:

	Hare	Hide
Stag	?, 2	?, 0
Hare	?, 1	?, 0

This is because 2 is greater than 0 and 1 is also greater than 0. Essentially, player 2 has no reason to hide if she knows player 1 would never hunt her. If we remove hide, the game reduces to the original stag hunt:

	Stag	Hare
Stag	3, 3	0, 2
Hare	2, 0	1, 1

From lesson 1.3, we know no more dominated strategies remain. To solve the game, we simply look for Nash equilibria as before. The two mutual best responses in pure strategies are <stag, stag> and <hare, hare>, so those are the two pure strategy Nash equilibria of this game. More pertinent to this topic, however, they are also the only two PSNE of the original game with human hunting and hiding.

<u>Nash Equilibrium and Iterated Elimination of Weakly Dominated Strategies</u>

Previously, we looked at this game:

	Left	Right
Up	0, 1	-4, 2
Middle	0, 3	3, 3
Down	-2, 2	3, -1

Middle weakly dominates both up and down for player 1. Depending on whether we chose to eliminate up or down first as a part of a sequence of iterated elimination of weakly dominated strategies, either <middle, left> was the only outcome that remained or <middle, right> was the only outcome that remained. So which of these is a pure strategy Nash equilibrium? Actually, *both* are.

As always, let's mark the best responses for both players. We begin by checking what happens if player 1 plays up:

	Left	Right
Up	?, 1	?, 2*

Since 2 is greater than 1, right is player 2's best response.

Now check on middle:

	Left	Right
Middle	?, 3*	?, 3*

If player 1 plays middle, player 2 is indifferent between left and right; both are worth 3 to her. As such, both are best responses.

Lastly, we go to down:

	Left	Right
Down	?, 2*	?, -1

Left is the best response here, as 2 is greater than -1.

From here, we move to player 1's best responses, beginning with player 2 moving left:

	Left
Up	0*, ?
Middle	0*, ?
Down	-2, ?

Up and middle are both best response to left; they both worth 0, while down yields -2.

The final situation to consider is player 2 moving right:

	Right
Up	-4, ?
Middle	3*, ?
Down	3*, ?

Again, player 1 has two best responses: middle and down.

Thus, the full game looks like this:

	Left	Right
Up	0*, 1	-4, 2*
Middle	0*, 3*	3*, 3*
Down	-2, 2	3*, -1

We see that both <middle, left> and <middle, right> are mutual best responses. Therefore, both are pure strategy Nash equilibria.

This game only offers a hint of why it is not a good idea to exclusively rely on weakly dominated strategies. After all, as long as you had considered every alternative sequence IESDS for this game, you would have found all the Nash equilibria. Unfortunately, weak dominance is much more finicky than that. In fact, even if there is only one path of elimination, IESWS may completely erase a Nash equilibrium.

Let's use this game as an example:

	Left	Right
Up	2, 3	4, 3
Down	3, 3	1, 1

The only dominance here is that left weakly dominates right for player 2:

	Left	Right
Up	?, 3	?, 3
Down	?, 3	?, 1

That is, if player 1 plays up, she is equally well off playing left or right, but she does strictly better playing left if he moves down.

If we remove right from the game, we are left with:

	Left
Up	2, 3
Down	3, 3

Since down now strictly dominates up for player 1, <down, left> is the solution that remains and it was also the *only* solution we can arrive at through iterated elimination of weakly dominated strategies.

Unfortunately, best responses tell a different story. Let's start with player 1's best response to player 2 playing left:

	Left
Up	2, ?
Down	3*, ?

As we just saw, it is down. Therefore, the 3 gets the asterisk.

Now check for player 1's best response to right:

	Right
Up	4*, ?
Down	1, ?

Player1's best response has flipped to up, as 4 is better than 1.

Switching perspectives, let's mark player 2's best response to player 1 moving up:

	Left	Right
Up	?, 3*	?, 3*

When player 1 goes up, player 2 is indifferent between left and right. As such, both are best responses.

Finally, let's check what happens when player 1 plays down:

	Left	Right
Down	?, 3*	?, 1

This time, left is the unique best response.

Bringing all of the best responses together, we get:

	Left	Right
Up	2, 3*	4*, 3*
Down	3*, 3*	1, 1

From this, we see there are two pure strategy Nash equilibria: <up, right> and <down, left>. There is no way we could have used iterated elimination of weakly dominated strategies to arrive at <up, right>, which is why it is so dangerous to use IEWDS by itself.

On the other hand, sometimes IEWDS turns out just fine. To wit, take a look at this game:

	Left	Center	Right
Up	2, 2	4, 2	4, 3
Middle	2, 4	5, 5	7, 3
Down	3, 4	3, 7	6, 6

To start, note that middle weakly dominates up:

	Left	Center	Right
Up	2, ?	4, ?	4, ?
Middle	2, ?	5, ?	7, ?

Middle always produces at least as great of a payoff as up. When player 2 plays center, middle beats up by 5 to 4. When she plays right, middle again wins by 7 to 4. However, both up and middle are worth 2 when she selects left. If we eliminated up as a strategy, the game reduces to this:

	Left	Center	Right
Middle	2, 4	5, 5	7, 3
Down	3, 4	3, 7	6, 6

And now we can observe that center strictly dominates left for player 2:

	Left	Center
Middle	?, 4	?, 5
Down	?, 4	?, 7

As the above image showcases, given that player 1 will not play up, if player 2 plays left, she will always earn 4 points. In contrast, she will earn 5 if player 1 plays middle and 7 if he plays down. So center strictly dominates left in the remaining game, and removing left yields the following:

	Center	Right
Middle	5, 5	7, 3
Down	3, 7	6, 6

Notice that center also strictly dominates right:

	Center	Right
Middle	?, 5	?, 3
Down	?, 7	?, 6

That is, 5 is greater than 3 and 7 is greater than 6. So we remove right from the game:

	Center
Middle	5, 5
Down	3, 7

Player 1 should pick middle and earn 5 rather than choose down and earn 3. Therefore, <middle, center> is the IEWDS solution and a pure strategy Nash equilibrium by extension. Unlike previous examples, however, it is also the *unique* pure strategy Nash equilibrium of the game. We can verify this through best responses. Let's begin with player 1's choice if player 2 goes left:

	Left
Up	2, ?
Middle	2, ?
Down	3*, ?

Since 3 is the largest number there, down gets the asterisk.

Next, check for player 1's response to center:

	Center
Up	4, ?
Middle	5*, ?
Down	3, ?

Again, there is a single best response: middle with 5, beating up with 4 and down with 3.

For the final best response for player 1, we check what happens when player 2 plays right:

	Right
Up	4, ?
Middle	7*, ?
Down	6, ?

Middle wins once more with a 7.

Switching to the other side, let's find player 2's best response if player 1 goes up:

	Left	Center	Right
Up	?, 2	?, 2	?, 3*

Right beats out left and center, as it is worth 3 points in response to up compared to the 2 for left and center.

Moving on, we now look at the best response to middle:

	Left	Center	Right
Middle	?, 4	?, 5*	?, 3

Here, the 5 points for center beat out the 4 for left and the 3 for right. So center is player 2's best response to middle.

Finally, we go to the best response to down:

	Left	Center	Right
Down	?, 4	?, 7*	?, 6

For the second time, center is the best response.

Throwing all of that together, the whole game looks like this:

	Left	Center	Right
Up	2, 2	4, 2	4, 3*
Middle	2, 4	5*, 5*	7*, 3
Down	3*, 4	3, 7*	6, 6

We see there is a single mutual best response: <middle, center>. Fortuitously, this was also the answer iterated elimination of weakly dominated strategies gave us. For once, IEWDS actually functioned nicely.

You might be wondering how you can tell whether IEWDS will work before you start solving the problem. Unfortunately, you cannot. It either will eliminate Nash equilibria or it will not, and you will not know which is the case until you check for best responses.

Simultaneous Strict and Weak Dominance

Occasionally, you will encounter a game that has both strictly and weakly dominated strategies in it. How should you go about solving the game? The most efficient answer is to begin by eliminating all of the strictly dominated strategies and only falling back on weak dominance when strict dominance cannot get you any further. The reason is simple: as soon as we eliminate a weakly dominated strategy—even just once—we may be eliminating some Nash equilibria.

To illustrate why, consider this game:

	Left	Right
Up	1, 1	2, -2
Middle	1, 0	1, -1
Down	0, 3	2, 2

Note that up weakly dominates middle for player 1:

	Left	Right
Up	1, ?	2, ?
Middle	1, ?	1, ?

That is, if player 2 plays left, player 1 is indifferent between up and middle; but if player 2 plays right, player 1 earns more from up than he does from middle. As such, up weakly dominates middle. Removing middle from the matrix yields this:

	Left	Right
Up	1, 1	2, -2
Down	0, 3	2, 2

Now left strictly dominates right for player 2:

	Left	Right
Up	?, 1	?, -2
Down	?, 3	?, 2

This is because 1 is greater than -2 and 3 is greater than 2. So if we remove the strictly dominated strategy, we are left with the following:

	Left
Up	1, 1
Down	0, 3

At this point, player 1 picks the strategy that gives him the greatest expected utility, so he selects up. Thus, we have arrived at <up, left> being a solution to

this game. Since a Nash equilibrium in a reduced game must also be a Nash equilibrium any of the previous games, <up, left> is a Nash equilibrium in the original game.

However, if we highlight the best responses of the original game, we find that the IEWDS process has eliminated another Nash equilibrium. To begin, consider player 1's best response to player 2 moving left:

	Left
Up	1*, ?
Middle	1*, ?
Down	0, ?

There are two: up and middle. Both provide 1 point, whereas down provides 0. As such, up and middle are best responses, but down is not.

Now let's check for his best response to right:

	Right
Up	2*, ?
Middle	1, ?
Down	2*, ?

Again, there are two: up and down. Middle loses out here by a score of 2 to 1.

Switching gears, we move to player 2's best response to up:

	Left	Right
Up	?, 1*	?, -2

Here, left is the best response, as 1 is greater than -2.

Moving on, let's find her best response to middle:

	Left	Right
Middle	?, 0*	?, -1

Again, left wins, so it earns the asterisk.

Finally, we go to down:

	Left	Right
Down	?, 3*	?, 2

For a third time, left is the best response.

Putting all of those best responses together, this is what the original game looks like:

	Left	Right
Up	1*, 1*	2*, -2
Middle	1*, 0*	1, -1
Down	0, 3*	2*, 2

Weak dominance has failed us once more: IEWDS produced an answer of <up, left>, but both <up, left> and <middle, left> are mutual best responses and are therefore pure strategy Nash equilibria. Utilizing weak dominance would have yielded only a partially correct answer here.

However, you might have noticed in the process of marking best responses that left strictly dominates right—regardless of what player 1 does, player 2 earns more by picking left than she does by picking right. So instead of beginning the elimination sequence with up weakly dominating middle, we could have begun by removing right. Had we done so, we would have been looking at this game:

	Left
Up	1, 1
Middle	1, 0
Down	0, 3

At this point, we up and middle both strictly dominate down, since 1 is greater than 0. Reducing the game one step further, we are left with only two outcomes:

	Left
Up	1, 1
Middle	1, 0

Since both up and middle are best responses to left, both <up, left> and <middle, left> are the IESDS solutions. And as we already know, they are also the only pure strategy Nash equilibria of the original game. Consequently, IESDS has preserved all the Nash equilibria.

Essentially, attacking the strictly dominated strategies first side-stepped the weak dominance problem. After we removed right from the game, there was no instance of weak dominance, only more cases of strict dominance. Unfortunately, this will not always be the case: sometimes after removing strictly dominated strategies you will still find weakly dominated strategies. At this point you will have to go through the best responses of the post-IESDS game as usual. However, when it does work, IESDS will save you a lot of time. As such, always pinpoint the strictly dominated strategies before you move on to weakly dominated strategies.

Lesson 1.5: Matching Pennies and Mixed Strategy Nash Equilibrium

Here is a very simple game. You and I each have a penny. Simultaneously, we choose whether to put our penny on the table with heads or tails facing up. If both of the pennies are heads or both of the pennies are tails (that is, they match), then you have to pay me a dollar. But if one is heads and the other is tails (that is, they do not match), then I have to pay you a dollar.

Suppose we value money equally. What is the optimal way to play this game? As usual, drawing a payoff matrix will help:

	Heads	Tails
Heads	1, -1	-1, 1
Tails	-1, 1	1, -1

This is an example of a strictly competitive or zero sum game; whatever player 1 wins, player 2 loses, and vice versa. There are plenty of real world examples that fit these payoffs. For example, on a soccer penalty kick, a kicker can kick left or right; the goalie can dive left or right. In American football, the offense can choose a running play or passing play; the defense can choose to protect against the pass or the run. In baseball, a pitcher can throw a fastball or a breaking ball; a batter can guess fastball or breaking ball. In war, an army can invade one of two cities; the defending side can adequately defend only one of them. In each of these games, one player is happy if their strategies match, while the other player prefers a mismatch.

Although all of these alternative examples are pertinent, let's stick to the matching pennies framework. Without even marking the best responses, it is clear that there are no pure strategy Nash equilibria in this game. Going through the logic, suppose player 1 always played heads. Then player 2 should play tails, since that will win her a dollar. But if player 2 is playing tails as a pure strategy, player 1 should play tails as well, as that will win him the dollar. But given that player 1 is playing tails, player 2 should switch over to heads. Yet if player 2 is playing heads, player 1 goes back to playing heads as well. The process now begins to loop, as now player 2's best response is to play tails. We have returned to our starting point.

If we analyzed the game by marking best responses, we would wind up with this:

	Heads	Tails
Heads	1*, -1	-1, 1*
Tails	-1, 1*	1*, -1

Unsurprisingly, every 1 has an asterisk and every -1 does not. Consequently, no outcome has an asterisk over each player's payoff, meaning there is no mutual best response in pure strategies. Therefore, there is no pure strategy Nash equilibrium.

Does that mean this game has no solution? Not nearly. As it turns out, *every* finite game has at least one Nash equilibrium. (John Forbes Nash, of *A Beautiful Mind* fame, proved this result, which is why we call this Nash's theorem. It is also why we search for "Nash" equilibria.) We define a game as being finite if the number of players is finite and the number of pure strategies each player has is finite. Here, there are two players, and each player has two pure strategies. Therefore, matching pennies is a finite game, so it must have a Nash equilibrium. We know matching pennies has no pure strategy Nash equilibria, so it must have an equilibrium in mixed strategies.

Mixed Strategies

Suppose I could read your mind. That is, as you were deciding whether to place your coin with heads or tails showing, I could perfectly anticipate what you were about to do. Is there any way you could minimize your losses?

The easiest solution would be to flip the coin and cover it with your hand as it is about to land on the table. If you did, my mind reading skills would be rendered harmless. No matter what I chose, my net expected gain from playing the game would be $0. Put differently, if I guessed heads every time, I would win $1 half the time and lose $1 half the time, for an average of $0. Likewise, if I guessed tails every time, I would win $1 half the time and lose $1 half the time, again for an average of $0.

Moreover, it is the case that if I flipped my coin in the same manner you do, I would still net $0. A quarter of the time they would both land on heads, and I would earn $1. Another quarter of the time they would both land on tails, and I would earn $1. In yet another quarter of the time, mine would land on heads and yours would land on tails, and I would lose $1. In the final quarter of the time, mine would land on tails and yours would land on heads, and I would again lose $1. On average, all of these cancel out, and my expected value of the game is equal to $0.

We could even extend this further and say that *any* probability distribution over playing heads and tails would net me $0 on average as long as you kept flipping your coin. For example, if I played heads 1/3 of the time and tails 2/3 of the time, I would earn $0. Likewise, if I played heads 7/16 of the time and tails 9/16 of the time, I would still be stuck earning $0. Effectively, your coin flipping

strategy is unbeatable.

From my perspective, if I cannot beat you, then I can just as easily flip my coin as well. As we just saw, I will make $0 from this game. But interestingly, if I am flipping my coin, then you cannot beat me for the same reason I could not beat you. So when we both flip our coins, it is impossible for either one of us to improve our payoffs. In other words, there are no profitable deviations from either of our strategies. Thus, this pair of coin flipping strategies is a mixed strategy Nash equilibrium.

Although this intuition is neat, limiting our possible mixed strategies to flipping only fair coins will not take us very far. Consider the following game:

	Left	Right
Up	3, -3	-2, 2
Down	-1, 1	0, 0

This game is very similar to matching pennies, except the values exchanged differ depending on the outcome; anywhere from 0 or 3 points exchanges hands depending on the outcome.

It is also still clear that there are no pure strategy Nash equilibria. If player 1 is playing up, then player 2 will want to play right and earn 2 points. But if player 2 is playing right, then player 1 will want to play down and earn 0 instead of -2. Yet if player 1 goes down, then player 2 can deviate to left and earn 1 point. However, that causes player 1 to switch to up and earn 3 points. Now player 2 wants to go right, and we have a new cycle.

Again, we can see this with the best responses marked:

	Left	Right
Up	3*, -3	-2, 2*
Down	-1, 1*	0*, 0

Matching pennies had the same pattern, with no outcome being a mutual best response. Since there are no pure strategy Nash equilibria, there must be one in mixed strategies.

But is merely flipping a coin sufficient? Unfortunately, no. Suppose player 1 flipped a fair coin and played up on heads and down on tails. Then player 2's expected utility for playing left is a weighted combination of her left column payoffs. We can write it as this:

$$EU_{left} = (.5)(-3) + (.5)(1) = -1$$

To see how we arrived at this equation, let's break down the game a step

further. We know that player 1 is playing both up and down some portion of the time, and we also know player 2 is always playing left. Since we only want to calculate player 2's payoff, we can block off all irrelevant information and end up with this:

	Left
Up	?, -3
Down	?, 1

Half the time, or (.5) of the time, player 1 will play up, and so player 2 will earn -3. That gives us the (.5)(-3) part of the equation. The second half follows similarly. Half the time, player 1 will play down, at which point player 2 earns 1. Thus, we get the (.5)(1) part. Summing those two parts together yields the answer of -1.

Let's do the same for player 2 playing right. Blocking out the irrelevant information gives us this:

	Right
Up	?, 2
Down	?, 0

Half of the time, she will earn 2. The other half of the time, she will earn 0. So her expected utility of playing right as a pure strategy in response to player 1 flipping a coin equals:

$$EU_{right} = (.5)(2) + (.5)(0) = 1$$

This is a problem. Clearly, if player 1 flips a coin, player 2 ought to play right as pure strategy since it generates a higher payoff for her. Moreover, player 2 would not want to randomize between left and right in response to player 1 flipping, as doing so feeds her more of the negative payoff from left and less of the positive payoff from right.

As such, player 2's best response is to play right. But if she's playing right, we already know player 1's best response is to play down as a pure strategy. Now player 1 has abandoned his mixture, and we are back to where we started, as the players cycle around their pure strategy best responses.

The Mixed Strategy Algorithm

Fortunately, there exists a specific algorithm we can use to find out how a player can induce his or her opponent to be indifferent between the opponent's

two pure strategies. On the down side, we now enter the computationally intensive part of game theory. Before, all we had to do was compare numbers and see which was larger. With mixed strategies, we have to introduce unknown variables and the necessary algebra to solve for them.

Let the lower case Greek letter sigma represent the probability that a player plays a particular pure strategy. For example, we use σ_{up} to be the probability that player 1 plays up. Using this notation, we can write player 2's expected utility of playing left as a pure strategy as a function of player 1's mixed strategy:

$$EU_{left} = (\sigma_{up})(-3) + (\sigma_{down})(1)$$

And player 2's expected utility of playing right as a pure strategy is:

$$EU_{right} = (\sigma_{up})(2) + (\sigma_{down})(0)$$

We are looking for a mixed strategy from player 1 that leaves player 2 indifferent between her pure strategies. In other words, we want to find a σ_{up} and σ_{down} such that:

$$EU_{left} = EU_{right}$$
$$(\sigma_{up})(-3) + (\sigma_{down})(1) = (\sigma_{up})(2) + (\sigma_{down})(0)$$

Although this may seem like a difficult task, note that σ_{up} and σ_{down} must sum to one. That is, since we know player 1 only has two strategies, he must play down whenever he does not play up. Therefore, we can use the substitution $\sigma_{down} = 1 – \sigma_{up}$ to rewrite that equation as:

$$(\sigma_{up})(-3) + (1 – \sigma_{up})(1) = (\sigma_{up})(2) + (1 – \sigma_{up})(0)$$

And from there, all we need to do is solve for σ_{up}, and we will have our answer:

$$-3\sigma_{up} + 1 - \sigma_{up} = 2\sigma_{up}$$
$$1 - 4\sigma_{up} = 2\sigma_{up}$$
$$6\sigma_{up} = 1$$
$$\sigma_{up} = 1/6$$

So if player 1 plays up with probability 1/6 and down with probability 5/6, then player 2 earns the same payoff for selecting either left or right as a pure strategy. Consequently, she also receives the same payoff if she chooses any mixture between left and right. This is important because player 2 will also have to play a mixed strategy in the Nash equilibrium of this game.

Knowing that, let's calculate the mixed strategy for player 2 that leaves player 1 indifferent between his two pure strategies. First, we need to ask ourselves what player 1's payoff for playing up is as a function of player 2's mixed strategy σ_{left}.

	Left	Right
Up	3, ?	-2, ?

With probability σ_{left}, player 1 earns 3; with probability $1 - \sigma_{left}$, he earns -2. We can write this as:

$$EU_{up} = (\sigma_{left})(3) + (1 - \sigma_{left})(-2)$$

Now we move on to player 1's payoff for playing down as a function of that same mixed strategy of player 2's:

	Left	Right
Down	-1, ?	0, ?

Like before, player 1 earns -1 with probability σ_{left} and 0 with probability $1 - \sigma_{left}$. As a formula:

$EU_{down} = (\sigma_{left})(-1) + (1 - \sigma_{left})(0)$

We need to find a mixed strategy for player 2 that leaves player 1 indifferent between his pure strategies. To do this, we set $EU_{up} = EU_{down}$ and solve for σ_{left}:

$EU_{up} = EU_{down}$
$EU_{up} = (\sigma_{left})(3) + (1 - \sigma_{left})(-2)$
$EU_{down} = (\sigma_{left})(-1) + (1 - \sigma_{left})(0)$
$(\sigma_{left})(3) + (1 - \sigma_{left})(-2) = (\sigma_{left})(-1) + (1 - \sigma_{left})(0)$
$3\sigma_{left} - 2 + 2\sigma_{left} = -\sigma_{left}$
$5\sigma_{left} - 2 = -\sigma_{left}$
$6\sigma_{left} = 2$
$\sigma_{left} = 1/3$

So if player 2 plays left with probability 1/3 and right with probability 2/3, then player 1 is indifferent between playing up and down as pure strategies. Moreover, he is indifferent between this and playing any mixture between up and down, as that mixture will still provide him the same payoff.

Connecting the mixed strategies of both players together, we see they are a best response to each other and therefore a Nash equilibrium. That is, if player 1 plays up with probability 1/6 and down with probability 5/6, then any strategy of player 2's will produce the same payoff, so any mixture between her two strategies is a best response. This includes mixing left with probability 1/3 and down with probability 2/3. Likewise, if player 2 plays left with probability 1/3 and right with probability 2/3, then any strategy of player 1's will produce the same payoff, so any mixture between his two strategies is a best response. This includes mixing up with probability 1/6 and down with probability 5/6. Since neither player can profitably change his or her strategy, those mixtures are a mixed strategy Nash equilibrium.

<u>How NOT to Write a Mixed Strategy Nash Equilibrium</u>

In the previous game, player 2 played left with probability 1/3 and right with probability 2/3. If we solved for her probability of playing left with a calculator, we would see something along the lines of 0.33333333. Likewise, the calculator

would generate 0.66666666 for her probability of playing right. Many students shorthand these to 0.33 and 0.67, implicitly meaning to write 1/3 and 2/3. But are 0.33 and 0.67 technically correct?

The answer is no. To see why, suppose player 2 played left with probability 0.33 and right with probability 0.67. For this to be a mixed strategy Nash equilibrium, player 1 must be indifferent between playing up and down. Knowing that, let's calculate player 1's expected utility of playing up and his expected utility of playing down. We will begin with up:

	Left	Right
Up	3, ?	-2, ?

If he plays up, he earns 3 with probability 0.33 and -2 with probability 0.67. As an equation:

$EU_{up} = (0.33)(3) + (0.67)(-2)$
$EU_{up} = 0.99 – 1.34$
$EU_{up} = -0.35$

Now let's find his expected utility if he plays down:

	Left	Right
Down	-1, ?	0, ?

This time, he earns -1 with probability 0.33 and 0 with probability 0.67. As an equation:

$EU_{down} = (0.33)(-1) + (0.67)(0)$
$EU_{down} = -0.33$

It should be apparent why this is not an equilibrium. If player 2 actually played left with probability 0.33 and right with probability 0.67, player 1's best response is to play down as a pure strategy, as -0.33 is greater than -0.35. The margin is slight, but down is better nonetheless. To induce indifference, the probabilities must be *exactly* 1/3 and *exactly* 2/3. Although 0.33 may seem innocuously different from 1/3, anything even slightly more or less than 1/3 breaks this all-important indifference.

Fortunately, there is a simple way to avoid this problem: write all of your answers as fractions. Decimals lose the precision for game theoretical calculations, but fractions always work perfectly. If you get into the habit of

using fractions exclusively, you will not lose points on an exam due on such a technicality.

Mixed Strategies in the Stag Hunt

Some games have both pure strategy Nash equilibria and mixed strategy Nash equilibria. The stag hunt from two lessons ago is one such game. Recall that both <stag, stag> and <hare, hare> are PSNE. However, there also exists a MSNE where both players mix between stag and hare. We can use the same algorithm from before to solve for it.

To differentiate between the players' strategies and for ease of exposition, let's replace the names of the players' strategies with up, down, left, and right, giving us this game:

	Left	Right
Up	3, 3	0, 2
Down	2, 0	1, 1

We start by finding player 2's expected utility for playing left as a pure strategy as a function of player 1's mixed strategy σ_{up}:

	Left
Up	?, 3
Down	?, 0

So with probability σ_{up}, player 2 earns 3. With complementary probability, she earns 0. As an equation:

$$EU_{left} = (\sigma_{up})(3) + (1 - \sigma_{up})(0)$$

Now we go to player 2's expected utility for playing right as a pure strategy as a function of that same mixed strategy σ_{up}:

	Right
Up	?, 2
Down	?, 1

Here, player 2 earns 2 with probability σ_{up} and 1 with probability $1 - \sigma_{up}$. Again, to write that as an equation:

$$EU_{right} = (\sigma_{up})(2) + (1 - \sigma_{up})(1)$$

To induce player 2's indifference, we set her expected utility for playing left equal to her expected utility for playing right and solve for σ_{up}:

$EU_{left} = EU_{right}$
$EU_{left} = (\sigma_{up})(3) + (1 - \sigma_{up})(0)$
$EU_{right} = (\sigma_{up})(2) + (1 - \sigma_{up})(1)$
$(\sigma_{up})(3) + (1 - \sigma_{up})(0) = (\sigma_{up})(2) + (1 - \sigma_{up})(1)$
$3\sigma_{up} = 2\sigma_{up} + 1 - \sigma_{up}$
$3\sigma_{up} = \sigma_{up} + 1$
$2\sigma_{up} = 1$
$\sigma_{up} = 1/2$

So if player 1 plays up and down each with probability 1/2, player 2 is indifferent between all combinations of left and right.

To complete the MSNE, we have to see what mixed strategy for player 2 leaves player 1 indifferent between his pure strategies. The process is the same, so we begin by looking at player 1's expected utility for playing up as a pure strategy as a function of player 2's mixed strategy σ_{left}:

	Left	Right
Up	3, ?	0, ?

He earns 3 with probability σ_{left} and 0 with probability $1 - \sigma_{left}$, or:

$EU_{up} = (\sigma_{left})(3) + (1 - \sigma_{left})(0)$

Now for down:

	Left	Right
Down	2, ?	1, ?

Here, he earns 2 with probability σ_{left} and 1 with probability $1 - \sigma_{left}$, or:

$EU_{down} = (\sigma_{left})(2) + (1 - \sigma_{left})(1)$

Combining these together yields the following:

$EU_{up} = EU_{down}$
$EU_{up} = (\sigma_{left})(3) + (1 - \sigma_{left})(0)$
$EU_{down} = (\sigma_{left})(2) + (1 - \sigma_{left})(1)$
$(\sigma_{left})(3) + (1 - \sigma_{left})(0) = (\sigma_{left})(2) + (1 - \sigma_{left})(1)$
$3\sigma_{left} = 2\sigma_{left} + 1 - \sigma_{left}$
$3\sigma_{left} = \sigma_{left} + 1$
$2\sigma_{left} = 1$
$\sigma_{left} = 1/2$

Thus, player 2 plays left and right each with probability 1/2 in the MSNE.

How Changing Payoffs Affects Mixed Strategy Nash Equilibria

When we solved the prisoner's dilemma, we saw that changing the payoffs did not matter much. Indeed, as long as the order of the numbers stayed the same—that is, we replaced the largest number from the first set of payoffs with the largest number from the second set of payoffs, we replaced the second largest number from the first set of payoffs with the second largest number from the second set of payoffs, and so forth—the equilibrium did not change. Regardless of the individual values, as long as the order of the numbers remained the same, confess still strictly dominated keep quiet, and thus <confess, confess> was the unique pure strategy Nash equilibrium of the game.

Mixed strategy Nash equilibrium is shakier. Although the MSNE will still exist as long as the order remains the same, even slight perturbations in relative values alters the exact probability distributions found in the MSNE.

To see this in action, let's look at a slightly altered stag hunt game:

	Left	Right
Up	4, 4	0, 2
Down	2, 0	1, 1

Let's solve for player 1's mixed strategy. As before, this requires us to define player 2's expected utility for each of her pure strategies as a function of player

1's mixed strategy σ_{up}. Let's begin her expected utility for left as a pure strategy:

	Left
Up	?, 4
Down	?, 0

With probability σ_{up}, she receives 4; and with probability $1 - \sigma_{up}$, she receives 0. Therefore, her expected utility is:

$$EU_{left} = (\sigma_{up})(4) + (1 - \sigma_{up})(0)$$

And now for player 2's expected utility for right:

	Right
Up	?, 2
Down	?, 1

This time, she earns 2 with probability σ_{up} and 1 with probability $1 - \sigma_{up}$. We can write this as:

$$EU_{right} = (\sigma_{up})(2) + (1 - \sigma_{up})(1)$$

Note that the expected utility for playing right has remained the same, but the expected utility for left has changed. As such, when we go through the last step of the process (setting EU_{left} equal to EU_{right} and solving for σ_{up}), we should expect a different mixed strategy. Sure enough, this is true:

$$EU_{left} = EU_{right}$$
$$EU_{left} = (\sigma_{up})(4) + (1 - \sigma_{up})(0)$$
$$EU_{right} = (\sigma_{up})(2) + (1 - \sigma_{up})(1)$$
$$(\sigma_{up})(4) + (1 - \sigma_{up})(0) = (\sigma_{up})(2) + (1 - \sigma_{up})(1)$$
$$4\sigma_{up} = 2\sigma_{up} + 1 - \sigma_{up}$$
$$4\sigma_{up} = \sigma_{up} + 1$$
$$3\sigma_{up} = 1$$
$$\sigma_{up} = 1/3$$

So player 1 plays up with probability 1/3 and down with probability 2/3 in the MSNE. The same steps will also show that player 2 makes the same change, now mixing left with probability 1/3 and right with probability 2/3. These mixtures are different from when the <stag, stag> outcome was worth 3 points for each player. Consequently, we must recalculate mixed strategy Nash equilibria every time a payoff changes.

Invalid Mixed Strategies

Not all games have a mixed strategy Nash equilibrium. Deadlock, for example, does not. To see this, let's try to use the algorithm to solve for an imaginary MSNE of the game. Recall that deadlock looked like this:

	Left	Right
Up	3, 3	4, 1
Down	1, 4	2, 2

Let's find the mixed strategy for player 2 that leaves player 1 indifferent between his pure strategies. We begin by isolating player 1's payoffs if he plays up as a pure strategy:

	Left	Right
Up	3, ?	4, ?

If player 2 plays σ_{left} as her mixed strategy and player 1 plays up as a pure strategy, player 1 earns 3 with probability σ_{left} and 4 with probability $1 - \sigma_{left}$. As an equation:

$$EU_{up} = (\sigma_{left})(3) + (1 - \sigma_{left})(4)$$

Now if player 1 plays down as a pure strategy:

	Left	Right
Down	1, ?	2, ?

This time, he earns 1 with probability σ_{left} and 2 with probability $1 - \sigma_{left}$. We write this as:

$$EU_{down} = (\sigma_{left})(1) + (1 - \sigma_{left})(2)$$

Let's attempt to set EU_{up} equal to EU_{down} and solve for σ_{left}:

$EU_{up} = EU_{down}$
$EU_{up} = (\sigma_{left})(3) + (1 - \sigma_{left})(4)$
$EU_{down} = (\sigma_{left})(1) + (1 - \sigma_{left})(2)$
$(\sigma_{left})(3) + (1 - \sigma_{left})(4) = (\sigma_{left})(1) + (1 - \sigma_{left})(2)$
$3\sigma_{left} + 4 - 4\sigma_{left} = \sigma_{left} + 2 - 2\sigma_{left}$
$-\sigma_{left} + 4 = -\sigma_{left} + 2$
$2 = 0$

In the process of solving for the mixed strategy, we eliminated σ_{left} from the equation in the last step and made the bold mathematical claim that 2 = 0. This is an absurd statement. Essentially, our algorithm is telling us that there is no mixed strategy that can make this work, and we can safely conclude that player 2 cannot mix in equilibrium. The same steps would also show that player 1 cannot mix in equilibrium. Thus, if neither player can mix in equilibrium, there are no mixed strategy Nash equilibria.

Other things can go wrong when we try to calculate a mixed strategy that cannot be played in equilibrium. Recall that the original prisoner's dilemma had the following payoff matrix:

	Keep Quiet	Confess
Keep Quiet	-1, -1	-12, 0
Confess	0, -12	-8, -8

Let's solve for player 1's mixed strategy. We begin by isolating player 2's payoffs for playing left (keep quiet):

	Keep Quiet
Keep Quiet	?, -1
Confess	?, -12

If player 2 plays left as a pure strategy, then she earns -1 with probability σ_{up} and -12 with probability $1 - \sigma_{up}$. Thus, her expected utility equals:

$EU_{left} = (\sigma_{up})(-1) + (1 - \sigma_{up})(-12)$

Now isolate player 2's payoffs for playing right (confess):

	Confess
Keep Quiet	?, 0
Confess	?, -8

Here, she earns 0 with probability σ_{up} and -8 with probability $1 - \sigma_{up}$. So her expected utility is:

$$EU_{right} = (\sigma_{up})(0) + (1 - \sigma_{up})(-8)$$

Let's try to solve for the mixed strategy played in the non-existent equilibrium:

$$EU_{left} = EU_{right}$$
$$EU_{left} = (\sigma_{up})(-1) + (1 - \sigma_{up})(-12)$$
$$EU_{right} = (\sigma_{up})(0) + (1 - \sigma_{up})(-8)$$
$$(\sigma_{up})(-1) + (1 - \sigma_{up})(-12) = (\sigma_{up})(0) + (1 - \sigma_{up})(-8)$$
$$-\sigma_{up} - 12 + 12\sigma_{up} = -8 + 8\sigma_{up}$$
$$-12 + 11\sigma_{up} = -8 + 8\sigma_{up}$$
$$3\sigma_{up} = 4$$
$$\sigma_{up} = 4/3$$

So, in this alleged MSNE, player 1 is supposed to play up with probability 4/3. We know that the probability he plays up is therefore 1 – 4/3, or -1/3. As such, he plays up with a probability greater than 100% and down a negative percentage of the time. Neither one of these things is sensible; all probabilities must be between 0 and 1. Once more, this is our algorithm's way of telling us that no such MSNE exists. From these two examples, we now know that a player cannot play a mixed strategy in equilibrium if the algorithm tells the player to do something impossible: play a strategy with an undefined probability or a probability greater than 1 or less than 0.

Unfortunately, if we show that one player cannot mix equilibrium, we still cannot eliminate the possibility that no MSNE exists. In particular, there are

games where one player plays a pure strategy while the other mixes. Some differentiate these as "partially mixed strategy Nash equilibria" because one player mixes and the other does not. Regardless, they still fall under the umbrella of MSNE. We will look at such a game in lesson 1.8.

Mixing and Dominance

Both the prisoner's dilemma and deadlock have a strictly dominated strategy. In addition, neither has a mixed strategy Nash equilibrium. Is this a coincidence?

Most certainly not. As it turns out, a strictly dominated strategy cannot be played with positive probability in a MSNE. Thinking about how mixed strategy Nash equilibria work reveals why. A requirement of Nash equilibrium is that no player has a profitable deviation from the strategies being played. So if a player plays a strictly dominated strategy, then some percentage of the time he ends up in an outcome which is strictly worse than if he had played the strictly dominant strategy instead. That implies that he can improve his payoff by playing the strictly dominant strategy whatever percentage of the time he would otherwise play the strictly dominated strategy. Thus, he has a profitable deviation whenever he plays the strictly dominated strategy.

To see how this operates, let's use the prisoner's dilemma as an example. Suppose player 1 could mix in equilibrium. Then some percentage of the time, he keeps quiet. If player 2 plays keep quiet as a pure strategy, then player 1 would regret playing keep quiet whenever his mixture told him to do so, as he could have played confess and spent less time in jail. Likewise, if player 2 plays confess as a pure strategy, then player 1 would again regret playing keep quiet whenever his mixture told him to do so for the same reason.

The logic is a little more complicated if player 2 were mixing. Suppose player 1 gets out his randomizing device, and it tells him to keep quiet. Given that, he knows player 2 confesses with probability $\sigma_{confess}$ and keeps quiet with probability $1 - \sigma_{confess}$, but he still does not know which until he actually reveals his choice to her. However, regardless of what she ends up choosing, player 1 could have done better by confessing. That is, for the $\sigma_{confess}$ amount of time, he would rather confess, and for the $1 - \sigma_{confess}$, he would still rather confess. The bottom line is that it makes no sense for him to play the strictly dominated strategy in a mixed strategy, so he does not.

However, weakly dominated strategies remain finicky, as they can be played in a MSNE. We will see such a case in lesson 1.8.

Lesson 1.6: Calculating Payoffs

When players play pure strategy Nash equilibria, their expected utilities are obvious. However, in a mixed strategy Nash equilibrium, we cannot simply look at an outcome and read a number to find a player's expected utility. This lesson shows how to calculate payoffs in MSNE using three commonly seen games: chicken, battle of the sexes, and pure coordination.

Chicken

Two testosterone-filled teenagers are on opposite ends of a one lane street and begin driving full speed toward one another. At the last possible moment, they must decide whether to swerve or continue going straight. If one continues while the other swerves, the one who swerves is a "chicken" while the other has proven his or her bravery. If both swerve, then neither can claim superiority. But if both continue, they crash straight into each other in a giant conflagration, which is the worst possible outcome for both players.

Although these teenagers' preferences may seem strange to us, we can still analyze their optimal behavior using the following utilities over outcomes:

	Continue	Swerve
Continue	-10, -10	2, -2
Swerve	-2, 2	0, 0

"Snowdrift" is an alternative title for this game. Under that framework, two drivers are stuck on the opposite ends of a snowy road, and they simultaneously decide whether to stay in their cars or shovel a passageway. An individual's most preferred outcome is to sit in his or her car while the other player shovels. That individual's next best outcome is for both players to shovel. The third best outcome is for that player to do the shoveling while the other stays inside. The worst outcome is for both players to stay in their cars, as the road never clears.

We will stick to chicken because it allows for the possibility of a fiery explosion and does not involve any depressing wintery weather. This game has two pure strategy Nash equilibria: <continue, swerve> and <swerve, continue>. Let's verify this by marking each player's best response, beginning with how player 1 should respond to player 2 continuing:

	Continue
Continue	-10, ?
Swerve	-2*, ?

Although both outcomes are undesirable to player 1, swerving minimizes his losses, so it earns the asterisk.

Next, suppose player 2 swerved:

	Swerve
Continue	2*, ?
Swerve	0, ?

Player 2 swerving opens up better outcomes for player 1. However, continuing is better than swerving here, meaning that player 1 wants to condition his behavior on what player 2 does.

The same is true for player 2. First, suppose player 1 continued:

	Continue	Swerve
Continue	?, -10	?, -2*

Then player 2 wants to swerve. But if he swerves:

	Continue	Swerve
Swerve	?, 2*	?, 0

Then she wants to continue. Hence, this is a game of mixed motives. Both players want to continue and force the other player to swerve. And, if they can convince the other player to back down, the game reaches a Nash equilibrium:

	Continue	Swerve
Continue	-10, -10	2*, -2*
Swerve	-2*, 2*	0, 0

However, each player cannot credibly threaten to continue if he or she knows the other will continue as well, as that leads to the disastrous <continue, continue> outcome. So perhaps there exists a MSNE as well. To find out, let's begin by renaming the strategies as directions to distinguish between each player's moves:

	Left	Right
Up	-10, -10	2, -2
Down	-2, 2	0, 0

First, let's solve for player 1's mixed strategy. Recall that we need to find player 2's expected utilities for each of her pure strategies as a function of player 1's mixture.

	Left
Up	?, -10
Down	?, 2

Player 2 earns -10 with probability σ_{up} and 2 with probability $1 - \sigma_{up}$. Therefore, her expected utility for left is:

$$EU_{left} = (\sigma_{up})(-10) + (1 - \sigma_{up})(2)$$

Now for right:

	Right
Up	?, -2
Down	?, 0

Here, player 2 earns -2 with probability σ_{up} and 0 with probability $1 - \sigma_{up}$, or:

$$EU_{right} = (\sigma_{up})(-2) + (1 - \sigma_{up})(0)$$

Setting these two expected utilities equal to each other allows us to solve for player 1's mixed strategy:

$$EU_{left} = EU_{right}$$
$$EU_{left} = (\sigma_{up})(-10) + (1 - \sigma_{up})(2)$$
$$EU_{right} = (\sigma_{up})(-2) + (1 - \sigma_{up})(0)$$
$$(\sigma_{up})(-10) + (1 - \sigma_{up})(2) = (\sigma_{up})(-2) + (1 - \sigma_{up})(0)$$
$$-10\sigma_{up} + 2 - 2\sigma_{up} = -2\sigma_{up}$$
$$-12\sigma_{up} + 2 = -2\sigma_{up}$$
$$10\sigma_{up} = 2$$
$$\sigma_{up} = 1/5$$

So player 1 goes up (continues) with probability 1/5 and goes down (swerves) with probability 4/5.

Using the same process for player 2's mixed strategy, we find an identical result. We begin with up as a pure strategy for player 1:

	Left	Right
Up	-10, ?	2, ?

So with probability σ_{left}, player 1 earns -10, and with probability $1 - \sigma_{left}$, he earns 2. As an equation:

$$EU_{up} = (\sigma_{left})(-10) + (1 - \sigma_{left})(2)$$

For down:

	Left	Right
Down	-2, ?	0, ?

Now he gets -2 with probability σ_{left} and 0 with probability $1 - \sigma_{left}$. We can write this as:

$$EU_{down} = (\sigma_{left})(-2) + (1 - \sigma_{left})(0)$$

Setting player 1's expected utility for up equal to his expected utility for down and solving for σ_{left} yields:

$$EU_{up} = EU_{down}$$
$$EU_{up} = (\sigma_{left})(-10) + (1 - \sigma_{left})(2)$$
$$EU_{down} = (\sigma_{left})(-2) + (1 - \sigma_{left})(0)$$
$$(\sigma_{left})(-10) + (1 - \sigma_{left})(2) = (\sigma_{left})(-2) + (1 - \sigma_{left})(0)$$
$$-10\sigma_{left} + 2 - 2\sigma_{left} = -2\sigma_{left}$$
$$-12\sigma_{left} + 2 = -2\sigma_{left}$$
$$10\sigma_{left} = 2$$
$$\sigma_{left} = 1/5$$

So in the MSNE, both players continue to drive straight with probability 1/5 and swerve with probability 4/5.

A natural question to ask is how the expected outcome of the MSNE compares to the outcomes in the PSNE. We can readily identify the value of the PSNE outcomes, since the payoff matrix explicitly tells us that the player who swerves gets -2 and the player who goes straight gets 2. However, all the outcomes occur with positive probability in the MSNE, which complicates matters. Indeed, we have to manually calculate them.

Fortunately, the process is easy. We simply take the probability that each outcome occurs in equilibrium and multiply it by the player's expected utility for that outcome. Let's isolate all of player 1's payoffs:

	Continue	Swerve
Continue	-10, ?	2, ?
Swerve	-2, ?	0, ?

We know that player 1 plays up with probability 1/5 and player 2 plays left with probability 4/5. So the probability player 1 earns his <continue, continue> payoff of -10 is simply the product of 1/5 and 4/5. Likewise, since he plays down with probability 4/5, the probability he earns his <swerve, continue> payoff of -2 is the product of 1/5 and 4/5. The probability <swerve, continue> occurs in the MSNE is the product of 4/5 and 1/5, so he earns 2 times that probability. Finally, the probability he earns 0 from the <swerve, swerve> outcome is the product of 4/5 and 4/5.

Player 1's expected utility is simply the sum of all of these products times the payoffs. In an equation form, solving for player 1's expected utility is routine:

$EU_1 = (1/5)(1/5)(-10) + (4/5)(1/5)(-2) + (1/5)(4/5)(2) + (4/5)(4/5)(0)$
$EU_1 = (1/25)(-10) + (4/25)(-2) + (4/25)(2) + (16/25)(0)$
$EU_1 = -10/25 – 8/25 + 8/25 + 0$
$EU_1 = -10/25$
$EU_1 = -2/5$

So if the players play the MSNE, player 1 expects to earn -2/5 on average. Note that he will not actually earn -2/5 when they play—he will either earn -10, -2, 0, or 2. But his *average* outcome is -2/5. This is similar to what you expect when you play a game like blackjack at a casino. You *expect* to lose a few cents per dollar you spend at the table, but on any given hand you either lose your entire bet or double it (or more if you hit a blackjack). Although there is no single outcome that ends with you receiving $0.98 or so of your original $1, it is nevertheless correct to say that $0.98 is your average outcome. When we find expected utilities of MSNE, we define the expectations analogously.

Player 2 has the same expected utility. We compute this the same way, beginning by isolating her payoffs:

	Continue	Swerve
Continue	?, -10	?, -2
Swerve	?, 2	?, 0

The probability each outcome occurs in equilibrium is the same as before. As such, it is just a matter of compiling these payoffs and their probabilities and then summing them together:

$EU_2 = (1/5)(1/5)(-10) + (1/5)(4/5)(2) + (4/5)(1/5)(-2) + (4/5)(4/5)(0)$
$EU_2 = (1/25)(-10) + (4/25)(2) + (4/25)(-2) + (16/25)(0)$
$EU_2 = -10/25 + 8/25 – 8/25 + 0$
$EU_2 = -10/25$
$EU_2 = -2/5$

As claimed, player 2's payoff is -2/5, just like player 1's.

Now that we know each player's expected utility in the MSNE, we can see the <swerve, swerve> outcome leaves both players better off, as 0 is greater than -2/5. However, as we saw with the prisoner's dilemma, such an outcome is inherently unstable, as one of the players could profitably deviate to continuing.

Battle of the Sexes

A man and a woman want to go on a date on a Friday evening in the 1980s. There are only two venues of entertainment in the city that night: a ballet and a fight. The woman wants to see the ballet. The man wants to see the fight. However, they both prefer being together than being alone, as they will have to immediately go home if the other does not show up at the location they choose. A simple cell phone call, text message, or email would simplify the coordination process, but the 1980s lacked those luxuries. As such, both must choose where to go simultaneously and without the ability to communicate with one another.

An alternative framework for this game pits two people deciding whether to go to a Bach or Stravinsky concert, but the central concepts and payoffs are the same. We can draw up the payoff matrix like this:

	Ballet	Fight
Ballet	1, 2	0, 0
Fight	0, 0	2, 1

Marking best responses to check for pure strategy Nash equilibrium is overkill here. If both go to the ballet, then neither has a profitable deviation, as they would become separated and earn 0 rather than some positive amount. So <ballet, ballet> is a Nash equilibrium. Likewise, if both go to the fight, then any deviation moves a player's payoff from a positive value to 0, so <fight, fight> is also a Nash equilibrium. Meanwhile, <ballet, fight> and <fight, ballet> are not Nash equilibria, as each player would rather go to wherever his or her date is than stick to his or her current strategy.

If we check for MSNE, we see one exists for this game. As always, to make things convenient, let's rename the strategies according to directions:

	Left	Right
Up	1, 2	0, 0
Down	0, 0	2, 1

We will solve for player 1's mixed strategy first, meaning we need isolate player 2's payoffs if she moves left:

	Left
Up	?, 2
Down	?, 0

So she earns 2 with probability σ_{up} and 0 with probability $1 - \sigma_{up}$. As an equation:

$$EU_{left} = (\sigma_{up})(2) + (1 - \sigma_{up})(0)$$

Now we look to her expected utility for playing right as a pure strategy:

	Right
Up	?, 0
Down	?, 1

Here, she earns 0 with probability σ_{up} and 1 with probability $1 - \sigma_{up}$. Once again, as an equation:

$$EU_{right} = (\sigma_{up})(0) + (1 - \sigma_{up})(1)$$

To find player 1's mixed strategy that makes player 2 indifferent, we set her expected utility for left equal to her expected utility for right and solve for σ_{up}:

$$EU_{left} = EU_{right}$$
$$EU_{left} = (\sigma_{up})(2) + (1 - \sigma_{up})(0)$$
$$EU_{right} = (\sigma_{up})(0) + (1 - \sigma_{up})(1)$$
$$(\sigma_{up})(2) + (1 - \sigma_{up})(0) = (\sigma_{up})(0) + (1 - \sigma_{up})(1)$$
$$2\sigma_{up} = 1 - \sigma_{up}$$
$$3\sigma_{up} = 1$$
$$\sigma_{up} = 1/3$$

So in the MSNE, player 1 goes to the ballet (up) with probability 1/3 and the fight (down) with probability 2/3.

Switching gears, let's solve for player 2's mixed strategy. To begin, consider player 1's payoffs for playing up as a function of that mixed strategy:

	Left	Right
Up	1, ?	0, ?

From this, we see that player 1 earns 1 with probability σ_{left} and 0 with probability $1 - \sigma_{left}$. So his expected utility for moving up is:

$$EU_{up} = (\sigma_{left})(1) + (1 - \sigma_{left})(0)$$

Now we must gather player 1's payoffs for down:

	Left	Right
Down	0, ?	2, ?

This time it is 0 with probability σ_{left} and 2 with probability $1 - \sigma_{left}$, or:

$$EU_{down} = (\sigma_{left})(0) + (1 - \sigma_{left})(2)$$

Finally, we set his expected utility for up equal to his expected utility for down and solve for σ_{left}:

$$EU_{up} = EU_{down}$$
$$EU_{up} = (\sigma_{left})(1) + (1 - \sigma_{left})(0)$$
$$EU_{down} = (\sigma_{left})(0) + (1 - \sigma_{left})(2)$$
$$(\sigma_{left})(1) + (1 - \sigma_{left})(0) = (\sigma_{left})(0) + (1 - \sigma_{left})(2)$$
$$\sigma_{left} = 2 - 2\sigma_{left}$$
$$3\sigma_{left} = 2$$
$$\sigma_{left} = 2/3$$

So player 2 goes to the ballet (left) with probability 2/3 and the fight (right) with probability 1/3.

In summary, each player goes to his or her preferred form of entertainment with probability 2/3 and his or her lesser preferred form of entertainment with probability 1/3. But what this information does not tell us is how inefficient of an outcome this is. To find out, we must calculate their expected utilities in the MSNE.

Let's isolate player 1's payoffs for each outcome:

	Ballet	Fight
Ballet	1, ?	0, ?
Fight	0, ?	2, ?

The <ballet, ballet> outcome occurs with probability 1/3 times 2/3. The <fight, ballet> outcome occurs with probability 2/3 times 2/3. The <ballet, fight> outcome occurs with probability 1/3 times 1/3. Finally, the <fight, fight> outcome occurs with probability 2/3 times 1/3. Multiplying these probabilities by their respective payoffs and then summing all of those payoffs together gives us the following:

$EU_1 = (1/3)(2/3)(1) + (2/3)(2/3)(0) + (1/3)(1/3)(0) + (2/3)(1/3)(2)$

$EU_1 = (2/9)(1) + (4/9)(0) + (1/9)(0) + (2/9)(2)$

$EU_1 = 2/9 + 4/9$

$EU_1 = 6/9$

$EU_1 = 2/3$

So player 1 earns a lowly 2/3 in the MSNE.

The same process reveals that player 2 earns 2/3 as well:

	Ballet	Fight
Ballet	?, 2	?, 0
Fight	?, 0	?, 1

Since the probabilities of each outcome remain the same, it is a simple matter of compiling the probabilities and payoffs:

$EU_1 = (1/3)(2/3)(2) + (2/3)(2/3)(0) + (1/3)(1/3)(0) + (2/3)(1/3)(1)$

$EU_1 = (2/9)(2) + (4/9)(0) + (1/9)(0) + (2/9)(1)$

$EU_1 = 4/9 + 2/9$
$EU_1 = 6/9$
$EU_1 = 2/3$

Why is this outcome so bizarre? Both the <ballet, fight> and <fight, ballet> outcomes represent coordination failure. They both occur with positive probability in the MSNE, accounting for 5/9 of the outcomes. That means the couple cannot go on their date more than half of the time if they mix, which drags down their payoffs. Indeed, each would be better off agreeing to meet at their lesser preferred form of entertainment and score their one point of utility, which beats the 2/3 of the point in the MSNE. Consequently, the MSNE is a rational but strange set of strategies. In turn, if players ever played the MSNE, we might wonder why they could not simply coordinate on one of the PSNE.

<u>Pure Coordination</u>

Mixed strategies can be even stranger in this regard. Consider the following game:

	Left	Right
Up	1, 1	0, 0
Down	0, 0	1, 1

We call this pure coordination. The players had conflicting motives in battle of the sexes; they wanted to be together, but they also wanted to see their preferred form of entertainment. In pure coordination, they only care about being together. One simple interpretation of this is choosing which side of the street to drive on. It does not really matter whether we all drive on the left side or all drive on the right side, as long as some of us do not drive on the left while others drive on the right.

Obviously, <up, left> and <down, right> are PSNE; any deviation from these sets of strategies changes a player's payoff from 1 to 0. However, there is also a mixed strategy Nash equilibrium. To solve for it, first consider player 2's payoffs for left as a function of player 1's mixed strategy σ_{up}:

	Left
Up	?, 1
Down	?, 0

She earns 1 with probability σ_{up} and 0 with probability $1 - \sigma_{up}$. In expected utility form:

$EU_{left} = (\sigma_{up})(1) + (1 - \sigma_{up})(0)$

Now for right:

	Right
Up	?, 0
Down	?, 1

This time, she earns 0 with probability σ_{up} and 1 with probability $1 - \sigma_{up}$. Or:

$EU_{right} = (\sigma_{up})(0) + (1 - \sigma_{up})(1)$

Putting these together, we solve for σ_{up}:

$EU_{left} = EU_{right}$
$EU_{left} = (\sigma_{up})(1) + (1 - \sigma_{up})(0)$
$EU_{right} = (\sigma_{up})(0) + (1 - \sigma_{up})(1)$
$(\sigma_{up})(1) + (1 - \sigma_{up})(0) = (\sigma_{up})(0) + (1 - \sigma_{up})(1)$
$\sigma_{up} = 1 - \sigma_{up}$
$2\sigma_{up} = 1$
$\sigma_{up} = 1/2$

So she mixes equally between left and right. Player 1 does the same with up and down. We can see this by starting with his expected utility for up as a function of σ_{left}:

	Left	Right
Up	1, ?	0, ?

Similar to what we saw with player 2, player 1 earns 1 with probability σ_{left} and 0 with probability $1 - \sigma_{left}$. We can write this as:

$EU_{up} = (\sigma_{left})(1) + (1 - \sigma_{left})(0)$

And for down:

	Left	Right
Down	0, ?	1, ?

Here, it is 0 with probability σ_{left} and 1 with probability $1 - \sigma_{left}$. Therefore, his expected utility for down equals:

$EU_{down} = (\sigma_{left})(0) + (1 - \sigma_{left})(1)$

Now we set EU_{up} equal to EU_{down} and solve for σ_{left}:

$EU_{up} = EU_{down}$
$EU_{up} = (\sigma_{left})(1) + (1 - \sigma_{left})(0)$
$EU_{down} = (\sigma_{left})(0) + (1 - \sigma_{left})(1)$
$(\sigma_{left})(1) + (1 - \sigma_{left})(0) = (\sigma_{left})(0) + (1 - \sigma_{left})(1)$
$\sigma_{left} = 1 - \sigma_{left}$
$2\sigma_{left} = 1$
$\sigma_{left} = 1/2$

Since each player plays both of his or her strategies with probability 1/2, each outcome occurs in the MSNE with probability 1/4. In two of these outcomes, the players earn 0. In the other two, they earn 1. As such, their expected utility in the MSNE is 1/2. This is strictly worse than either of the PSNE, which is problematic for the players.

One way to escape the inefficient mixed strategy Nash equilibria in pure coordination and battle of the sexes is to follow social norms and laws. Driving on the road in the United States is very easy because a law tells us to drive on the right side, and that is an efficient Nash equilibrium. In battle of the sexes, perhaps the couple had a rule of thumb that the man chooses where to go on Fridays and the woman chooses where to go on Saturdays. If that were the case, they would only need to look at a calendar to coordinate even if they could not directly communicate. Thus, these strange MSNE help us interpret the usefulness of these types of coordination rules.

A Short Cut with Zero-Sum Games

In chicken and battle of the sexes, each player's payoff in the MSNE is equal

to his or her opponent's. However, unless the game is as symmetrical as the ones above, that will usually not be the case. To see an example where the payoffs differ, refer back the modified form of matching pennies discussed in lesson 1.5:

	Left	Right
Up	3, -3	-2, 2
Down	-1, 1	0, 0

Begin by recalling that player 1 mixes with probability 1/6 on up and 5/6 on down, while player 2 mixes with probability 1/3 on left and 2/3 on right. Therefore, <up, left> occurs with probability 1/6 times 1/3; <down, left> occurs with probability 5/6 times 1/3; <up, right> occurs with probability 1/6 times 2/3; and <down, right> occurs with probability 5/6 times 2/3.

Let's focus on player 1's payoffs and use these probabilities to calculate his expected utility:

	Left	Right
Up	3, ?	-2, ?
Down	-1, ?	0, ?

So his expected utility equals:

$EU_1 = (1/6)(1/3)(3) + (5/6)(1/3)(-1) + (1/6)(2/3)(-2) + (5/6)(2/3)(0)$

$EU_1 = (1/18)(3) + (5/18)(-1) + (2/18)(-2) + (10/18)(0)$

$EU_1 = 3/18 – 5/18 – 4/18$

$EU_1 = -6/18$

$EU_1 = -1/3$

Rather than calculating player 2's expected utility as usual, recall that this game is zero sum: every time a player gains some amount, the other player loses that amount. In other words, if we sum the payoffs for each individual outcome, all add up to 0. Given that player 1's gain is player 2's loss and vice versa, if player 1's expected utility equals -1/3, then player 2's expected utility must be the negative of that payoff, or 1/3.

In summary, we have learned two things from this lesson. First, calculating payoffs in mixed strategy Nash equilibria of zero sum games is easy, since you functionally calculate both players' payoffs by just doing one player's. And second, the players' payoffs need not be equal in MSNE; "equilibrium" only refers to the stability of certain strategies, not any sort of balance in the players' payoffs.

Lesson 1.7: Strictly Dominant Mixed Strategies

Consider the following game:

	Left	Right
Up	3, -1	-1, 1
Middle	0, 0	0, 0
Down	-1, 2	2, -1

Specifically, note that middle is not dominated by player 1's other pure strategies. To see this, let's first compare middle to up:

	Left	Right
Up	3, ?	-1, ?
Middle	0, ?	0, ?

If player 2 played left, player 1 would rather go up and earn 3, which is better than the 0 he would earn for middle. But if player 2 played right, player 1 would now rather go middle, as 0 is greater than the -1 he earns for up. So up does not dominate middle.

Likewise, down does not dominate middle either:

	Left	Right
Middle	0, ?	0, ?
Down	-1, ?	2, ?

If player 2 plays left, then player 1 is better off going middle; but if she goes right, then he ought to go down. So down does not dominate middle.

Combining these two pieces of information together, we now know that no pure strategy dominates middle. Nevertheless, middle is strictly dominated. How? Well, if a mixture of two pure strategies strictly dominates a third strategy, then that third strategy is strictly dominated as normal.

To see how middle is strictly dominated, consider a mixed strategy from player 1 where he plays up with probability 1/2 and down with probability 1/2. Let's calculate his expected utility if player 2 plays left:

	Left
Up	3, ?
Down	-1, ?

Using this matrix for guidance, we know that he will earn 3 with probability 1/2 and -1 with probability 1/2. As an equation:

$EU_1 = (1/2)(3) + (1/2)(-1)$

$EU_1 = 3/2 – 1/2$

$EU_1 = 2/2$

$EU_1 = 1$

Note that his expected utility for this mixed strategy is greater than the 0 that he would earn if he played middle as pure strategy.

Now suppose player 2 played right and player 1 stuck to this mixed strategy:

	Right
Up	-1, ?
Down	2, ?

Now he earns -1 with probability 1/2 and 2 with probability 2. Therefore, his expected utility equals:

$EU_1 = (1/2)(-1) + (1/2)(2)$

$EU_1 = -1/2 + 2/2$

$EU_1 = 1/2$

Again, this expected utility is greater than the 0 he would earn if he played middle in response to player 2 playing right. That means regardless of player 2's choice, player 1 would be better off playing this mixture between up and down than playing middle as a pure strategy. Thus, middle is strictly dominated.

Other mixtures between up and down also strictly dominate middle. For example, up with probability 49/100 and down with probability 51/100 or up with probability 51/100 and down with probability 49/100 will both produce a strictly better outcome than middle regardless of what player 2 does. However, this is superfluous information. Once we know that a mixture—any mixture at all—of some pure strategies strictly dominates another pure strategy, we can immediately remove that dominated pure strategy just as we have done in the past.

Specifically, that means we can reduce the original game to this far less intimidating form:

	Left	Right
Up	3, -1	-1, 1
Down	-1, 2	2, -1

It is easy to see that there are no pure strategy Nash equilibria here. For the <up, left> outcome, player 2 can deviate to right and improve from -1 to 1. For the <up, right> outcome, player 1 can deviate to down and improve from -1 to 2. For the <down, right> outcome, player 2 can deviate to left and improve from -1 to 2. Finally, for the <down, left> outcome, player 1 can deviate to up and

improve from -1 to 3.

As such, we look to the mixed strategy algorithm to provide a solution. Let's start with player 1's mixed strategy, meaning we need to find player 2's payoff for playing left as a pure strategy in response to that mixture:

	Left
Up	?, -1
Down	?, 2

So with probability σ_{up}, player 2 earns -1; and with probability $1 - \sigma_{up}$, she earns 2. Therefore, her expected utility is:

$$EU_{left} = (\sigma_{up})(-1) + (1 - \sigma_{up})(2)$$

Next, we need to find her expected utility for right against that same mixed strategy:

	Right
Up	?, 1
Down	?, -1

Here, she earns 1 with probability σ_{up} and -1 with probability $1 - \sigma_{up}$. As an equation:

$$EU_{right} = (\sigma_{up})(1) + (1 - \sigma_{up})(-1)$$

To finish, we set the expected utility for left equal to the expected utility for right and solve for σ_{up}:

$$EU_{left} = EU_{right}$$
$$EU_{left} = (\sigma_{up})(-1) + (1 - \sigma_{up})(2)$$
$$EU_{right} = (\sigma_{up})(1) + (1 - \sigma_{up})(-1)$$
$$(\sigma_{up})(-1) + (1 - \sigma_{up})(2) = (\sigma_{up})(1) + (1 - \sigma_{up})(-1)$$
$$-\sigma_{up} + 2 - 2\sigma_{up} = \sigma_{up} - 1 + \sigma_{up}$$
$$-3\sigma_{up} + 2 = 2\sigma_{up} - 1$$
$$5\sigma_{up} = 3$$

$\sigma_{up} = 3/5$

So player 1 plays up with probability 3/5 and down with probability 2/5 in the MSNE.

Let's move on to player 2's mixed strategy. We begin by isolating player 1's payoffs for up in response to her mixture:

	Left	Right
Up	3, ?	-1, ?

Player 1 earns 3 with probability σ_{left} and -1 with probability $1 - \sigma_{left}$. Or:

$$EU_{up} = (\sigma_{left})(3) + (1 - \sigma_{left})(-1)$$

Now for down:

	Left	Right
Down	-1, ?	2, ?

Here, he earns -1 with probability σ_{left} and 2 with probability $1 - \sigma_{left}$. As an expected utility equation:

$$EU_{down} = (\sigma_{left})(-1) + (1 - \sigma_{left})(2)$$

Finally, to solve for the exact mixture, we set his expected utility for up equal to his expected utility for down:

$$EU_{up} = EU_{down}$$
$$EU_{up} = (\sigma_{left})(3) + (1 - \sigma_{left})(-1)$$
$$EU_{down} = (\sigma_{left})(-1) + (1 - \sigma_{left})(2)$$
$$(\sigma_{left})(3) + (1 - \sigma_{left})(-1) = (\sigma_{left})(-1) + (1 - \sigma_{left})(2)$$
$$3\sigma_{left} - 1 + \sigma_{left} = -\sigma_{left} + 2 - 2\sigma_{left}$$
$$4\sigma_{left} - 1 = -3\sigma_{left} + 2$$
$$7\sigma_{left} = 3$$
$$\sigma_{left} = 3/7$$

So in the MSNE, player 1 plays up with probability 3/5 and down with probability 2/5 while player 2 plays left with probability 3/7 and right with probability 4/7.

Although strict dominance guarantees that player 1 will not have profitable deviation to middle, let's verify that this is the case. Specifically, note that player 1 earns a guaranteed payoff of 0 for selecting middle:

	Left	Right
Middle	0, ?	0, ?

Effectively, he can force a payoff of 0 for himself, as player 2's strategy becomes irrelevant for him if he selects middle. Consequently, if player 1 cannot profitably deviate from the MSNE to playing middle as a pure strategy, it must be the case that his expected utility from the MSNE is greater than or equal to 0.

To check, let's calculate the probability of each outcome occurring in the MSNE. Given the mixtures up with probability 3/5, down with probability 2/5, left with probability 3/7, and right with probability 4/7, the likelihood that each outcome occurs is simply the respective probabilities multiplied together. That is, <up, left> occurs with probability 3/5 times 3/7; <down, left> occurs with probability 2/5 times 3/7; <up, right> occurs with probability 3/5 times 4/7; and <down, right> occurs with probability 2/5 times 4/7.

All we need to do is match up those probabilities with player 1's payoffs for each of the outcomes:

	Left	Right
Up	3, ?	-1, ?
Down	-1, ?	2, ?

We now combine this information together as a formula:

$EU_1 = (3/5)(3/7)(3) + (2/5)(3/7)(-1) + (3/5)(4/7)(-1) + (2/5)(4/7)(2)$

$EU_1 = (9/35)(3) + (6/35)(-1) + (12/35)(-1) + (8/35)(2)$

$EU_1 = 27/35 – 6/35 – 12/35 + 16/35$

$EU_1 =25/35$

$EU_1 = 5/7$

And, sure enough, his expected utility of 5/7 in the MSNE is better than 0 he would receive for middle as a pure strategy.

Mixed Dominance and IESDS

We can use mixed dominance in a sequence of iterated elimination of strictly dominated strategies as normal. To see this, we will use this game:

	Left	Center	Right
Up	-3, 6	9, 1	0, 2
Middle	3, -4	2, 4	4, 1
Down	4, 7	3, 2	-3, 2

Here, no pure strategy strictly dominates right for player 2. We can see isolating left versus right and center versus right. First, let's compare right with left:

	Left	Right
Up	?, 6	?, 2
Middle	?, -4	?, 1
Down	?, 7	?, 2

Between these two, left is better than right for player 2 if player 1 plays up or down; right is worth 2 for her in each of these cases, while left is worth 6 if he moves up and 7 if he moves down. However, right beats left if player 1 chooses middle, as 1 is greater than -4. As such, left does not strictly dominate right.

Next, we compare center to right:

	Center	Right
Up	?, 1	?, 2
Middle	?, 4	?, 1
Down	?, 2	?, 2

Now right is better if player 1 chooses up, by a 2 to 1 margin. In contrast, if player 1 goes middle, player 2's center move earns her 4, which is greater than the 1 she would earn by choosing right. Finally, if player 1 goes down, she earns 2 regardless of which of these two strategies she chooses. So center does not strictly dominate right.

However, a mixture between left and center does. Let's calculate the expected utility for player 2 if she plays left with probability 1/4 and center with probability 3/4. First, suppose player 1 plays up as a pure strategy:

	Left	Center
Up	?, 6	?, 1

So player 2 earns 6 with probability 1/4 and 1 with probability 3/4. Therefore, her expected utility equals:

$EU_2 = (1/4)(6) + (3/4)(1)$

$EU_2 = 6/4 + 3/4$
$EU_2 = 9/4$

In contrast, player 2 earns 2 if she plays right in this situation. Since 9/4 is greater 2, the mixture provides a greater payoff for her than playing right as pure strategy if player 1 plays up.

Now we go to player 1 playing middle as a pure strategy:

	Left	Center
Middle	?, -4	?, 4

Here, player 2 earns -4 with probability 1/4 and 4 with probability 3/4. As such, her expected utility equals:

$EU_2 = (1/4)(-4) + (3/4)(4)$
$EU_2 = -4/4 + 12/4$
$EU_2 = 8/4$
$EU_2 = 2$

Meanwhile, if she had played right in response to player 1 playing middle, she would have earned 1. Again, the mixture provides a better payoff with 2.

Finally, we move on to player 1 playing down:

	Left	Center
Down	?, 7	?, 2

In this case, player 2 earns 7 with probability 1/4 and 2 with probability 3/4. It follows that her expected utility equals:

$EU_2 = (1/4)(7) + (3/4)(2)$
$EU_2 = 7/4 + 6/4$
$EU_2 = 13/4$

Playing right as pure strategy would have earned her 2. Because 13/4 is greater than 2, player 2 again does better with the mixed strategy.

Indeed, she always does better with the mixture between left and center than

she does with right as a pure strategy, so the mixed strategy strictly dominates right. As such, we can remove it from the matrix, thereby reducing the game to the following:

	Left	Center
Up	-3, 6	9, 1
Middle	3, -4	2, 4
Down	4, 7	3, 2

From here, we can continue our IESDS process as normal. Note that down now strictly dominates middle for player 1:

	Left	Center
Middle	3, ?	2, ?
Down	4, ?	3, ?

That is, 4 is greater than 3 if player 2 plays left and 3 is greater than 2 if she plays center. Since down strictly dominates middle, we can remove middle from the matrix and leave the following:

	Left	Center
Up	-3, 6	9, 1
Down	4, 7	3, 2

Left now strictly dominates center for player 2:

	Left	Center
Up	?, 6	?, 1
Down	?, 7	?, 2

If player 1 moves up, left beats center 6 to 1. If he moves down, left wins 7 to 2. Either way, player 2 is better off playing left than she is playing center. Therefore, we can eliminate center, which leaves us with just two outcomes:

	Left
Up	-3, 6
Down	4, 7

From here, player 1 simply picks the best outcome for himself. Since 4 is greater than -3, he moves down. Thus, through iterated elimination of strictly dominated strategies, <down, left> is the only Nash equilibrium of this game.

Overall, strict dominance in mixed strategies can be frustrating to work with —there are many combinations of pure strategies and an infinite range of mixtures between those strategies. Consequently, it takes effort to locate such strictly dominant mixed strategies. However, the payoff is ultimately worth it, as we can simplify games a great deal when we do find them.

Lesson 1.8: The Odd Rule and Infinitely Many Equilibria

Let's think back to some of the games we have solved and how many Nash equilibria they had. The prisoner's dilemma and deadlock each had one pure strategy Nash equilibrium that we could find through strict dominance. Many other games we have solved through IESDS also had one. Matching pennies had one in mixed strategies. The stag hunt, chicken, and battle of the sexes each had three: two in pure strategies and one in mixed strategies.

Is it a coincidence that all of these games have an odd number of equilibria? Actually, it is not—it is rare to see a game with an even number or infinite number of equilibria. However, some quirky games do not follow this odd rule of thumb, and our old friend weak dominance frequently claims responsibility. Consider this game, which has exactly two Nash equilibria:

	Left	Right
Up	1, 1	0, 0
Down	0, 0	0, 0

Up weakly dominates down:

	Left	Right
Up	1, ?	0, ?
Down	0, ?	0, ?

That is, if player 2 plays left, then up is better than down; but if player 2 plays right, up and down are equally good.

Likewise, left weakly dominates right:

	Left	Right
Up	?, 1	?, 0
Down	?, 0	?, 0

Here, if player 1 plays up, left is better than right; but if player 1 plays down, left and right are equally good.

That means <up, left> is a pure strategy Nash equilibrium. This should not come as a surprise given that <up, left> provides the greatest payoff for both players. However, <down, right> is also a PSNE. To see this, suppose player 1 played down. Then no matter what player 2 does, she will earn 0. As such, right is a best response to down. Similarly, if player 2 is playing right, player 1 will earn 0 whether he plays up or down. Therefore, down is a best response to right. That means if the players choose <down, right> they cannot individually improve their payoffs to a value greater than zero. Thus, <down, right> is a mutual best response and a Nash equilibrium.

Clearly <down, left> and <up, right> are not. In the case of <down, left>, player 1 can profitably deviate to up and increase his payoff from 0 to 1; likewise, in the case of <up, right>, player 2 can profitably deviate to left and

increase her payoff from 0 to 1. So if this game only has two Nash equilibria, then there must be no mixed strategy Nash equilibrium.

To verify the absence of a MSNE, suppose player 1 mixed with probability σ_{up}. Then player 2's payoff for playing left is:

$$EU_{left} = (\sigma_{up})(1) + (1 - \sigma_{up})(0)$$
$$EU_{left} = \sigma_{up}$$

And her payoff for playing right is:

$$EU_{right} = (\sigma_{up})(0) + (1 - \sigma_{up})(0)$$
$$EU_{right} = 0$$

Consequently, if $\sigma_{up} > 0$ (that is, as long as player 1 is truly mixing and not playing down with certainty), then player 2's best response to any such mixture is to play left as pure strategy. The reason is simple—if she plays left, there is some chance she will receive 1, but she is guaranteed to earn 0 if she plays down.

Player 1's best response to a mixture from player 2 works the same way. Suppose player 2 mixed with probability σ_{left}. Then player 1's payoff for playing up is:

$$EU_{up} = (\sigma_{left})(1) + (1 - \sigma_{left})(0)$$
$$EU_{up} = \sigma_{left}$$

And his payoff for playing down is:

$$EU_{down} = (\sigma_{left})(0) + (1 - \sigma_{left})(0)$$
$$EU_{down} = 0$$

So just as before, if player 2 puts positive probability on left and right, then player 1's best response is to always play up. As a result, the players cannot mix, thus making this a rare game that has an even number of equilibria.

Infinitely Many Equilibria

Games can also have infinitely many equilibria. We will look at two different ways this can occur. First, consider this game:

	Left	Right
Up	2, 2	9, 0
Down	2, 3	5, -1

To solve it, we should begin by noting that left strictly dominates right for player 2:

	Left	Right
Up	?, 2	?, 0
Down	?, 3	?, -1

If player 1 plays up, left is better than right for player 2 by a 2 to 0 margin. If he plays down, left beats right 3 to -1. So regardless of what player 1 does, player 2 ought to move left. Thus, through iterated elimination of strictly dominated strategies, we can reduce the game to this:

	Left
Up	2, 2
Down	2, 3

Knowing that player 2 will play left, player 1 earns 2 no matter what he does. Therefore, both up and down are pure strategy responses to left. In turn, both <up, left> and <down, left> are pure strategy Nash equilibria.

While that should be obvious, a subtle implication here is that *any* mixture between up and down is also a best response to left. To see this, consider player 1's expected utility for any mixed strategy. We know player 2 plays left, so player 1's payoff is as follows:

$$EU_1 = (\sigma_{up})(2) + (1 - \sigma_{up})(2)$$

That is, he earns 2 with probability σ_{up} (whatever percentage of the time he chooses to play up) and 2 with probability $1 - \sigma_{up}$ (whatever percentage of the

time he chooses to play down). However, this formula quickly simplifies to 2:

$$EU_1 = (\sigma_{up})(2) + (1 - \sigma_{up})(2)$$
$$EU_1 = 2\sigma_{up} + 2 - 2\sigma_{up}$$
$$EU_1 = 2$$

Since this is equal to his payoff for playing up or down pure strategy, all of these mixtures are also best responses to left. As such, in equilibrium, player 1 can play up as pure strategy, down as a pure strategy, or mix in any fashion between those two. There are infinitely many of such mixtures; therefore, this game has infinitely many equilibria. Although this falls under the MSNE umbrella, we can be very specific and call this a partially mixed strategy Nash equilibrium, as player 1 mixes but player 2 does not.

Another interesting thing to note is that player 1 plays a weakly dominated strategy with positive probability in the MSNE. To see this, let's focus on player 1's payoffs in isolation:

	Left	Right
Up	2, ?	9, ?
Down	2, ?	5, ?

As we have just seen, if player 1 plays left, then both up and down are equally as good. But putting right back into the matrix reveals that up is better for player 1 in that case. Consequently, up weakly dominates down. However, if we eliminated down, the <down, left> PSNE and the set of MSNE where player 1 mixes between up and down with any probability both disappear. In case the point about weak dominance was not already clear, this game provides another example about why we have to be extremely careful about eliminating weakly dominated strategies.

Moving on, there is a second type of games with infinitely many equilibria. The previous example was fairly trivial, because player 1's best response to player 2's strategy was uninteresting—he earned 2 no matter what he did. No change in his strategy would make player 2 want to change her strategy. But that is not the case in all games. Recall back to this game, which we saw in lesson 1.4:

	Left	Right
Up	2, 3	4, 3
Down	3, 3	1, 1

After marking best responses, we saw that <down, left> and <up, right> were both pure strategy Nash equilibria:

	Left	Right
Up	2, 3*	4*, 3*
Down	3*, 3*	1, 1

Note that both left and right are best responses to up for player 2. Consequently, as long as player 1 plays up as a pure strategy, player 2 is free to mix between left and right in any combination. Given that, we should wonder what sorts of mixed strategies employed by player 2 would induce player 1 to play up over down.

Unlike the previous game, this requires a few calculations. First, we need to know player 1's payoff for playing up as a function of player 2's mixed strategy:

	Left	Right
Up	2, ?	4, ?

So player 1 earns 2 with probability σ_{left} and 4 with probability $1 - \sigma_{left}$. We can write that as this equation:

$$EU_{up} = (2)(\sigma_{left}) + (4)(1 - \sigma_{left})$$

Moving on to down:

	Left	Right
Down	3, ?	1, ?

Now player 1 earns 3 with probability σ_{left} and 1 with probability $1 - \sigma_{left}$. As an equation:

$$EU_{down} = (3)(\sigma_{left}) + (1)(1 - \sigma_{left})$$

Remember that a strategy is a best response if it provides at least as good of a payoff as any other alternative. Thus, playing up is a best response to player 2's mixed strategy if the expected utility of playing up is greater than or equal to player 1's expected utility for playing down. Bringing together the previous two equations and solving for the inequality yields the following:

$EU_{up} \geq EU_{down}$
$EU_{up} = (2)(\sigma_{left}) + (4)(1 - \sigma_{left})$
$EU_{down} = (3)(\sigma_{left}) + (1)(1 - \sigma_{left})$
$(2)(\sigma_{left}) + (4)(1 - \sigma_{left}) \geq (3)(\sigma_{left}) + (1)(1 - \sigma_{left})$
$2\sigma_{left} + 4 - 4\sigma_{left} \geq 3\sigma_{left} + 1 - \sigma_{left}$
$-2\sigma_{left} + 4 \geq 2\sigma_{left} + 1$
$4\sigma_{left} \leq 3$
$\sigma_{left} \leq 3/4$

From this, we know that up is a best response for player 1 if player 2 plays left less than or equal to 3/4 of the time (and thus plays right at least 1/4 of the time). Essentially, such a mixture successfully deters player 1 from choosing down, thereby securing player 2 her payoff of 3. And since player 2 earns 3 regardless of what she does if player 1 plays up, she too is playing a best response to his strategy. Therefore, there are infinitely many Nash equilibria: <down, left>, <up, right>, and infinitely many in mixed strategies where player 1 plays up and player 2 plays left with probability no greater than 3/4.

Once more, weak dominance is at fault; left weakly dominates right for player 1:

	Left	Right
Up	?, 3	?, 3
Down	?, 3	?, 1

We already know player 2 is indifferent between left and right if player 1 plays up, but left beats right for her if he moves down. Absent that weak dominance, the infinitely many equilibria disappear.

Here is a similar game:

	Left	Right
Up	3, 1	0, 0
Down	2, 2	2, 2

There are two pure strategy Nash equilibria: <up, left> and <down, right>. We can verify this by marking best responses. To begin, suppose player 1 went up:

	Left	Right
Up	?, 1*	?, 0

Since 1 is greater than 0, left is player 2's best response to up. On the other hand, if player 1 went down:

	Left	Right
Down	?, 2*	?, 2*

Now player 2 earns 2 regardless of what she does. As such, both left and right are best responses.

Let's move on to player 1's best responses. To start, suppose player 2 played left:

	Left
Up	3*, ?
Down	2, ?

Then up is the best response, as 3 is greater than 2. Finally, consider what happens if player 2 plays right:

	Right
Up	0, ?
Down	2*, ?

Since 2 is greater than 0, down is the best response here.

Putting all of that together, the original game looks like this:

	Left	Right
Up	3*, 1*	0, 0
Down	2, 2*	2*, 2*

Since <up, left> and <down, right> have asterisks for both players payoffs, they are mutual best responses and therefore pure strategy Nash equilibrium.

However, notice that left and right are both best responses for player 2 if player 1 plays down. As such, provided that player 1 selects down as a pure strategy, player 2 can mix freely between left and right. Consequently, we need to check if down is a best response for player 1 to any of these mixtures from player 2.

To do so, we need to write out player 1's expected utility for each of his pure strategies as a function of player 2's mixed strategy. Let's begin with up:

	Left	Right
Up	3, ?	0, ?

In this case, player 1 earns 3 with probability σ_{left} and 0 with probability $1 - \sigma_{left}$. As an equation:

$$EU_{up} = (3)(\sigma_{left}) + (0)(1 - \sigma_{left})$$
$$EU_{up} = 3\sigma_{left}$$

Next, we must check down:

	Left	Right
Down	2, ?	2, ?

Here, he earns 2 with probability σ_{left} and 2 with probability $1 - \sigma_{left}$. Unsurprisingly, that simplifies to just 2. So regardless of player 2's mixed strategy, player 1 earns 2 if he selects down:

$EU_{down} = (2)(\sigma_{left}) + (2)(1 - \sigma_{left})$
$EU_{down} = 2\sigma_{left} + 2 - 2\sigma_{left}$
$EU_{down} = 2$

Recall that we are looking for all mixed strategies of player 2 that make down a best response for player 1. That is, we need to know for what values of σ_{left} is player 1's expected utility for down greater than or equal to his expected utility for up. A little bit of algebra gives us the answer:

$EU_{up} \leq EU_{down}$
$EU_{up} = 3\sigma_{left}$
$EU_{down} = 2$
$3\sigma_{left} \leq 2$
$\sigma_{left} \leq 2/3$

Ergo, this game has infinitely many mixed strategy Nash equilibria in which player 1 plays down as a pure strategy and player 2 plays left with probability no greater than 2/3 and plays right with the remaining probability.

Once more, weak dominance is at work:

	Left	Right
Up	?, 1	?, 0
Down	?, 2	?, 2

Left weakly dominates right; if player 1 moves down, then player 2 earns 2 regardless of her move. But if player 1 chooses up, then the 1 for left is greater than the 0 for right.

<u>Why Does the Odd Rule Matter?</u>

As you solve games on your own, count the number of equilibria you find. If

a game turns up an even number of an infinite number of equilibria, give it a second look. Your answer might be correct, but it is worth double checking given the rarity of such games.

Final Thought

This book has quickly explained all of the tools you will need to solve introductory level questions about matrix form games. Good luck in applying them.

And always remember: weak dominance will be the bane of your existence.

About the Author

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